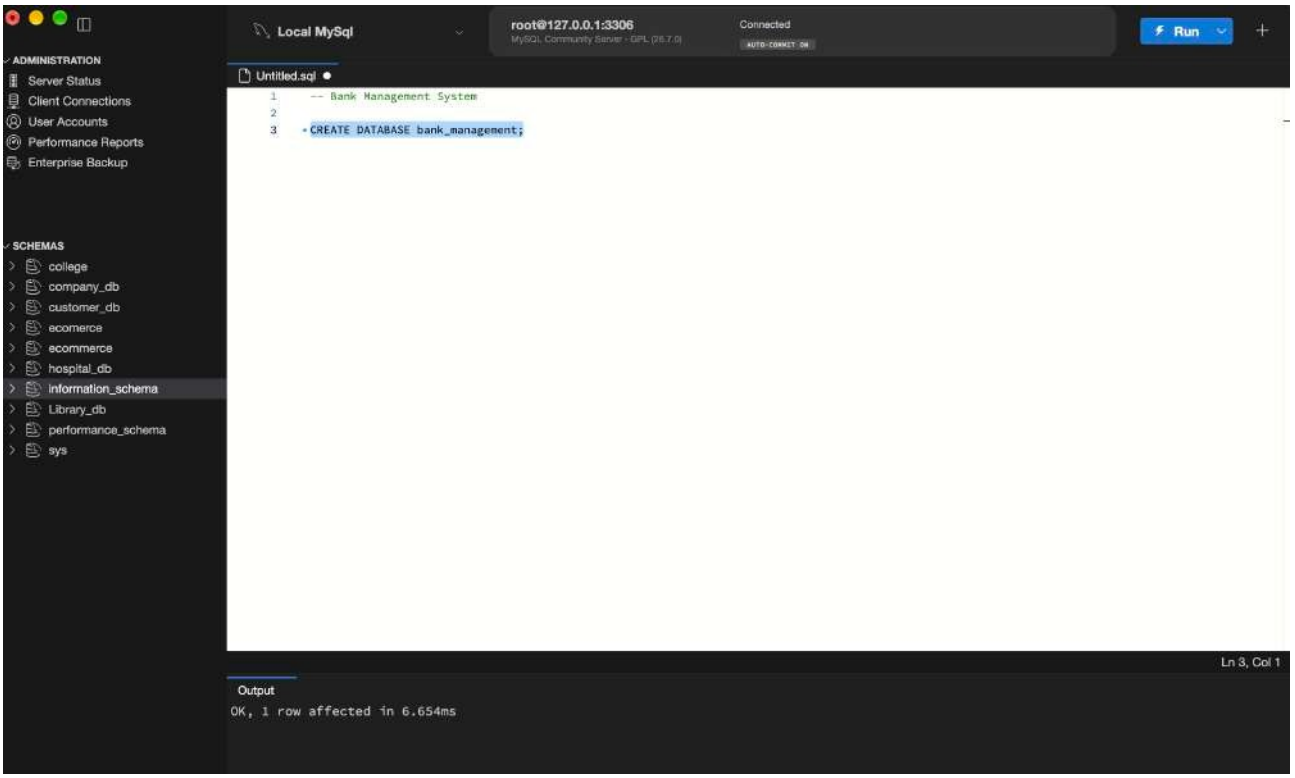
AWS end to end data engineering pipeline

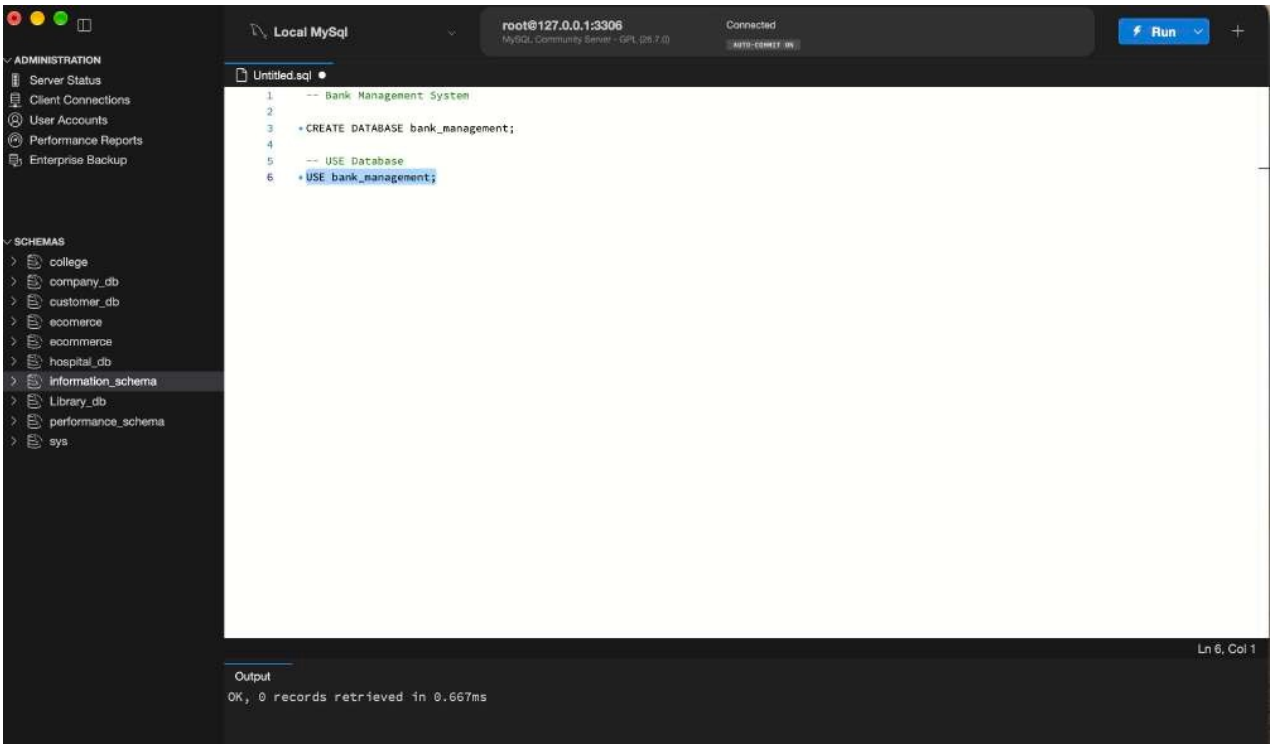
Name : Gorlamandala Nanda Sai
Roll Number: 23BTRCL042
College Name : JAIN DEEMED TO BE UNIVERSITY
Batch : 2023-2027
HCLID : HCLTFP2262836

SQL Practice: Bank Management System

— Creating the database



— Using the database



— Creating a tables

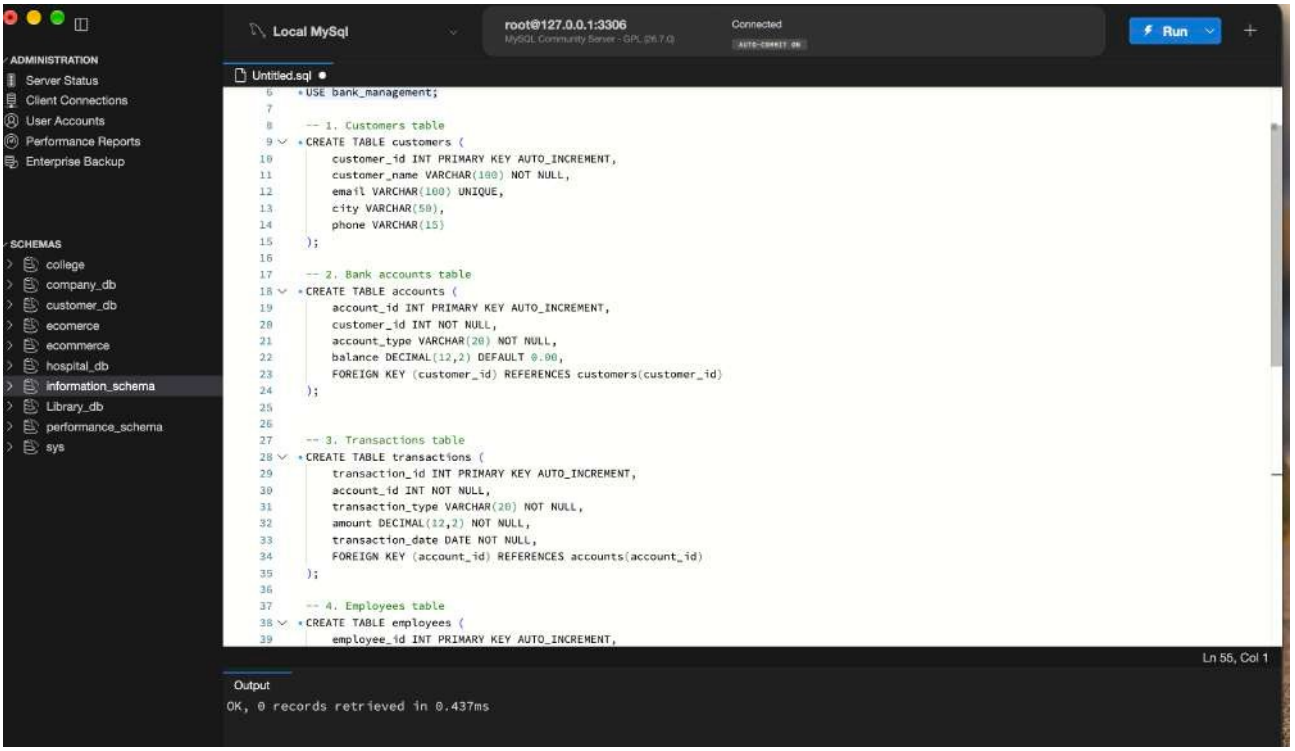
1.Customers table

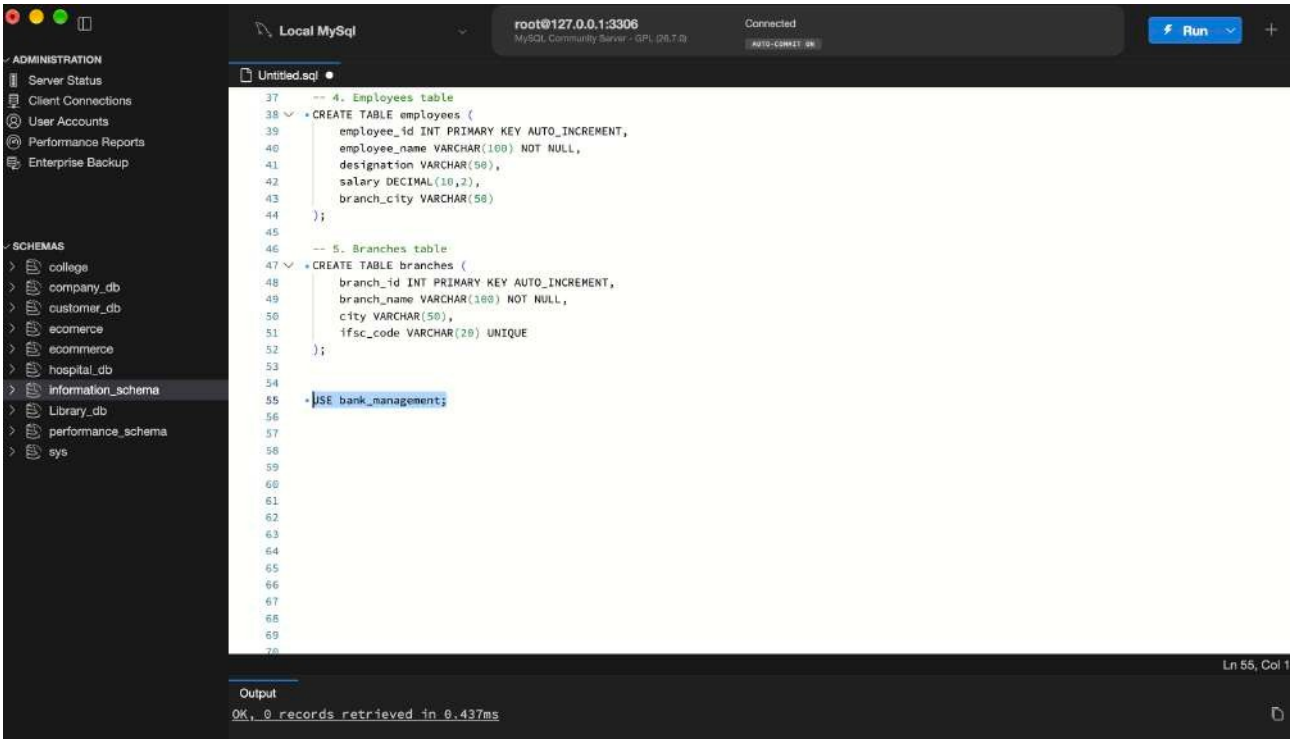
2.Bank accounts table

3.Transactions table

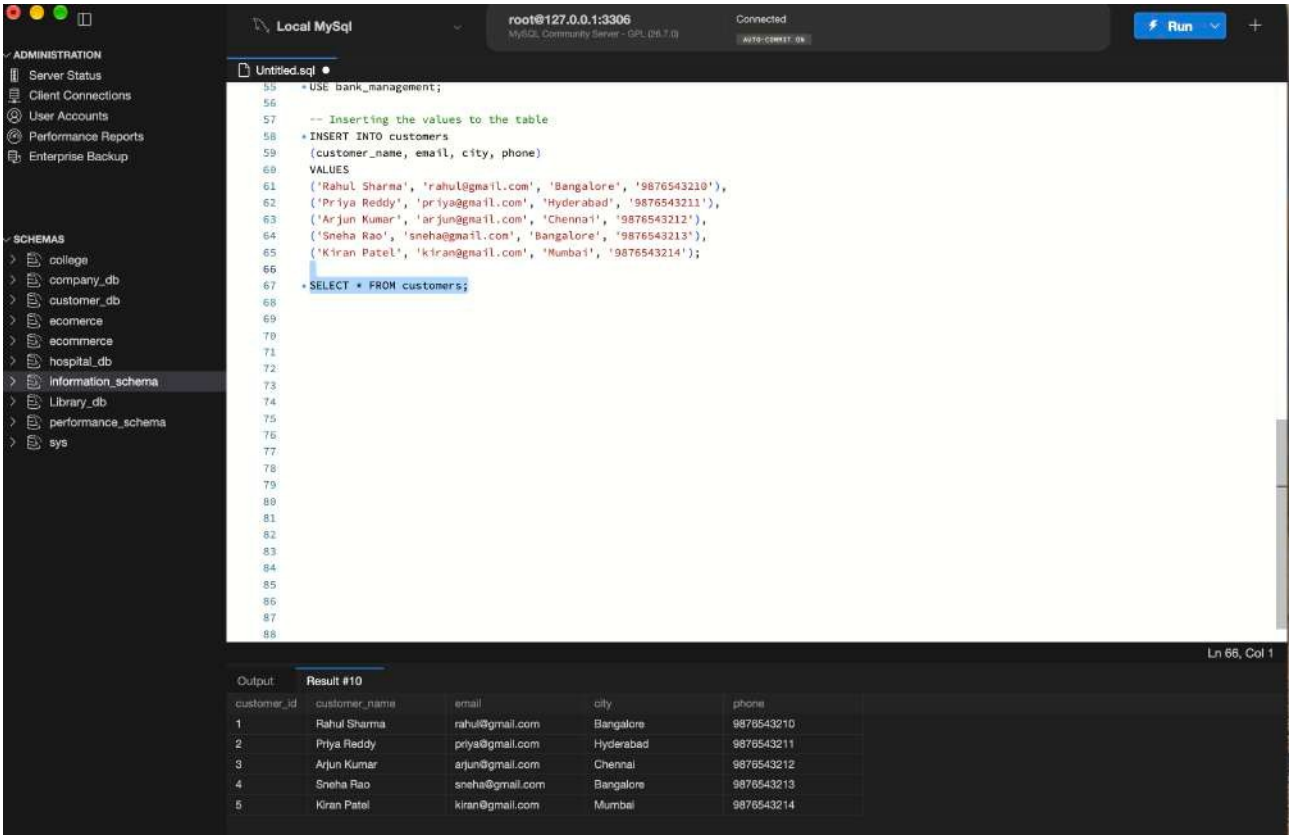
4.Employees table

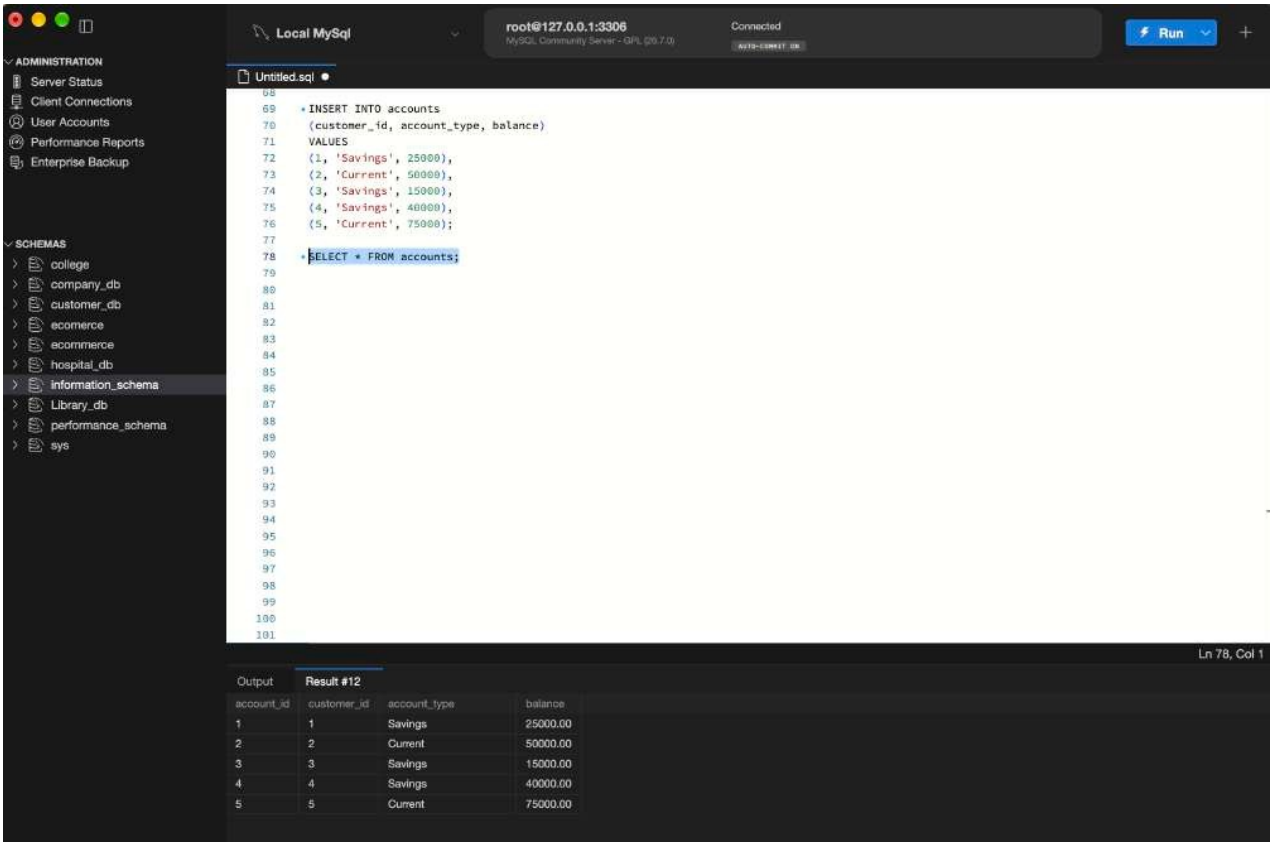
5.Branches table

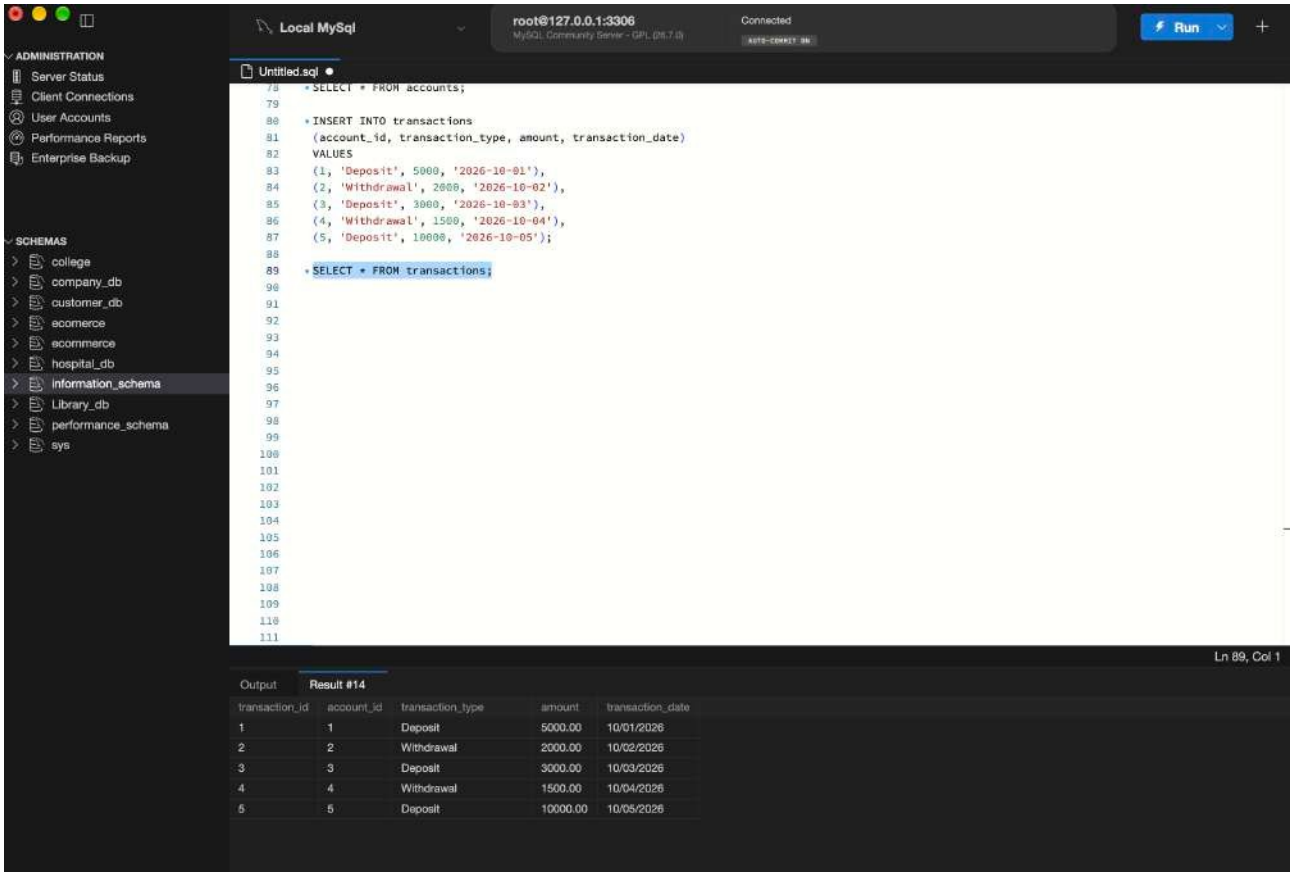


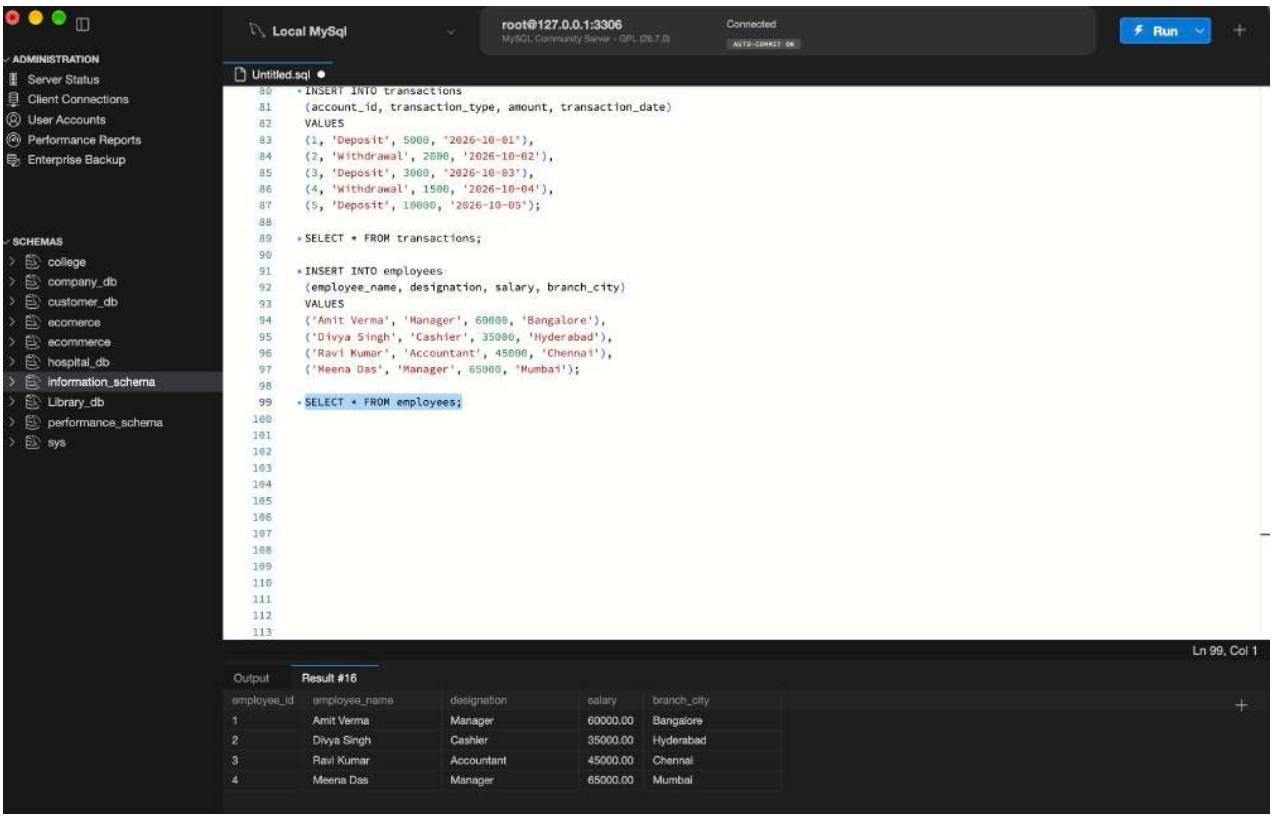


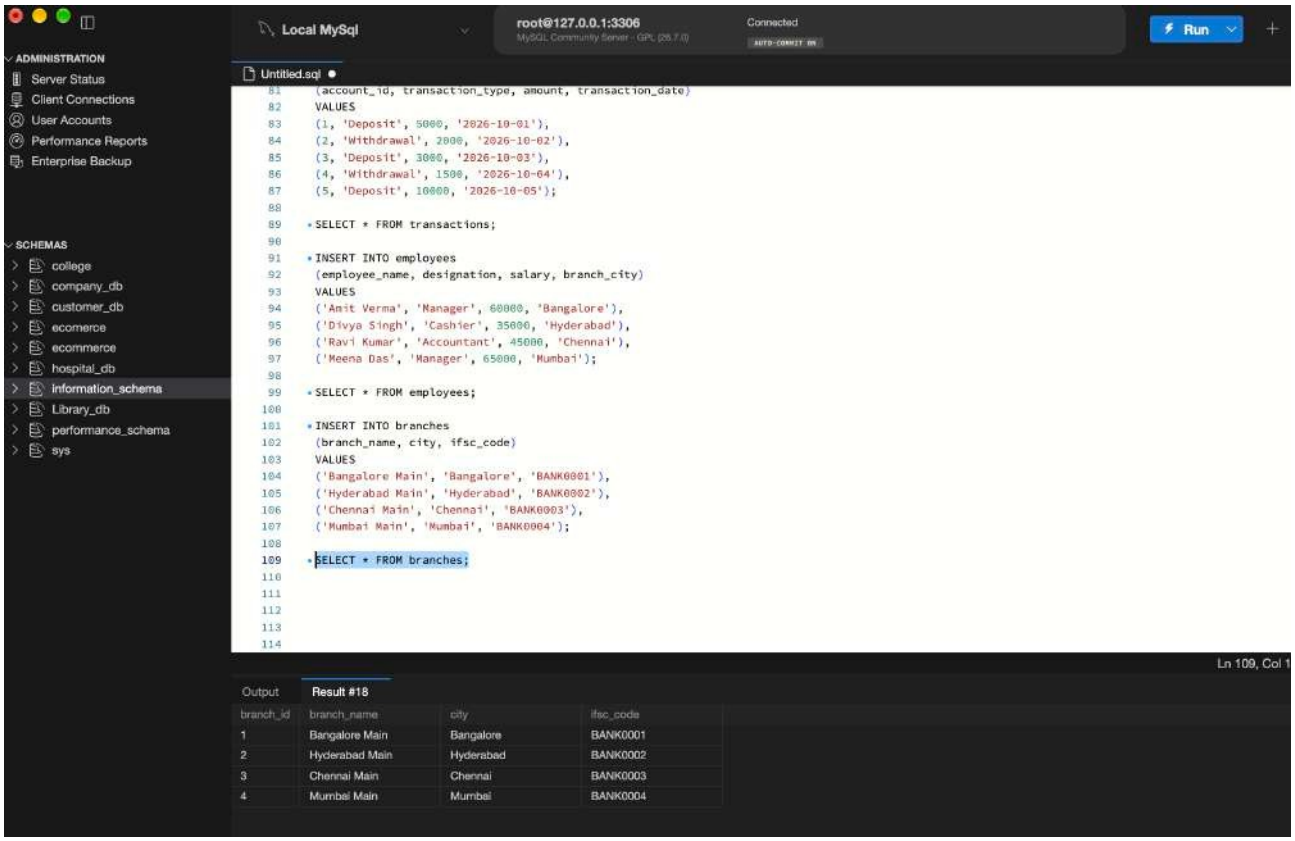
—Inserting the values to the tables






Local MySql
root@127.0.0.1:3306
MySQL Community Server - GPL (8.7.0)
Connected
AUTO-COMMIT ON
Run
ADMINISTRATION
Server Status
Client Connections
User Accounts
Performance Reports
Enterprise Backup
SCHEMAS
college
company_db
customer_db
ecomerce
ecommerce
hospital_db
information_schema
Library_db
performance_schema
sys
Untitled.sql
SELECT * FROM accounts;
INSERT INTO transactions
(account_id, transaction_type, amount, transaction_date)
VALUES
(1, 'Deposit', 5000, '2026-10-01'),
(2, 'Withdrawal', 2000, '2026-10-02'),
(3, 'Deposit', 3000, '2026-10-03'),
(4, 'Withdrawal', 1500, '2026-10-04'),
(5, 'Deposit', 10000, '2026-10-05');
SELECT * FROM transactions;
Ln 89, Col 1
Output
Result #14
transaction_id
account_id
transaction_type
amount
transaction_date
1
1
Deposit
5000.00
10/01/2026
2
2
Withdrawal
2000.00
10/02/2026
3
3
Deposit
3000.00
10/03/2026
4
4
Withdrawal
1500.00
10/04/2026
5
5
Deposit
10000.00
10/05/2026


Local MySql
root@127.0.0.1:3306
MySQL Community Server - GPL (8.7.0)
Connected
AUTO-COMMIT ON
Run
ADMINISTRATION
Server Status
Client Connections
User Accounts
Performance Reports
Enterprise Backup
SCHEMAS
college
company_db
customer_db
ecomerce
ecommerce
hospital_db
information_schema
Library_db
performance_schema
sys
Untitled.sql
INSERT INTO transactions
(account_id, transaction_type, amount, transaction_date)
VALUES
(1, 'Deposit', 5000, '2026-10-01'),
(2, 'Withdrawal', 2000, '2026-10-02'),
(3, 'Deposit', 3000, '2026-10-03'),
(4, 'Withdrawal', 1500, '2026-10-04'),
(5, 'Deposit', 10000, '2026-10-05');
SELECT * FROM transactions;
INSERT INTO employees
(employee_name, designation, salary, branch_city)
VALUES
('Amit Verma', 'Manager', 60000, 'Bangalore'),
('Divya Singh', 'Cashier', 35000, 'Hyderabad'),
('Ravi Kumar', 'Accountant', 45000, 'Chennai'),
('Meena Das', 'Manager', 65000, 'Mumbai');
SELECT * FROM employees;
Ln 99, Col 1
Output
Result #16
employee_id
employee_name
designation
salary
branch_city
1
Amit Verma
Manager
60000.00
Bangalore
2
Divya Singh
Cashier
35000.00
Hyderabad
3
Ravi Kumar
Accountant
45000.00
Chennai
4
Meena Das
Manager
65000.00
Mumbai



— Verify the tables

```
-- Verify your database

SHOW TABLES;

SELECT * FROM customers;
SELECT * FROM accounts;
SELECT * FROM transactions;
SELECT * FROM employees;
SELECT * FROM branches;
```

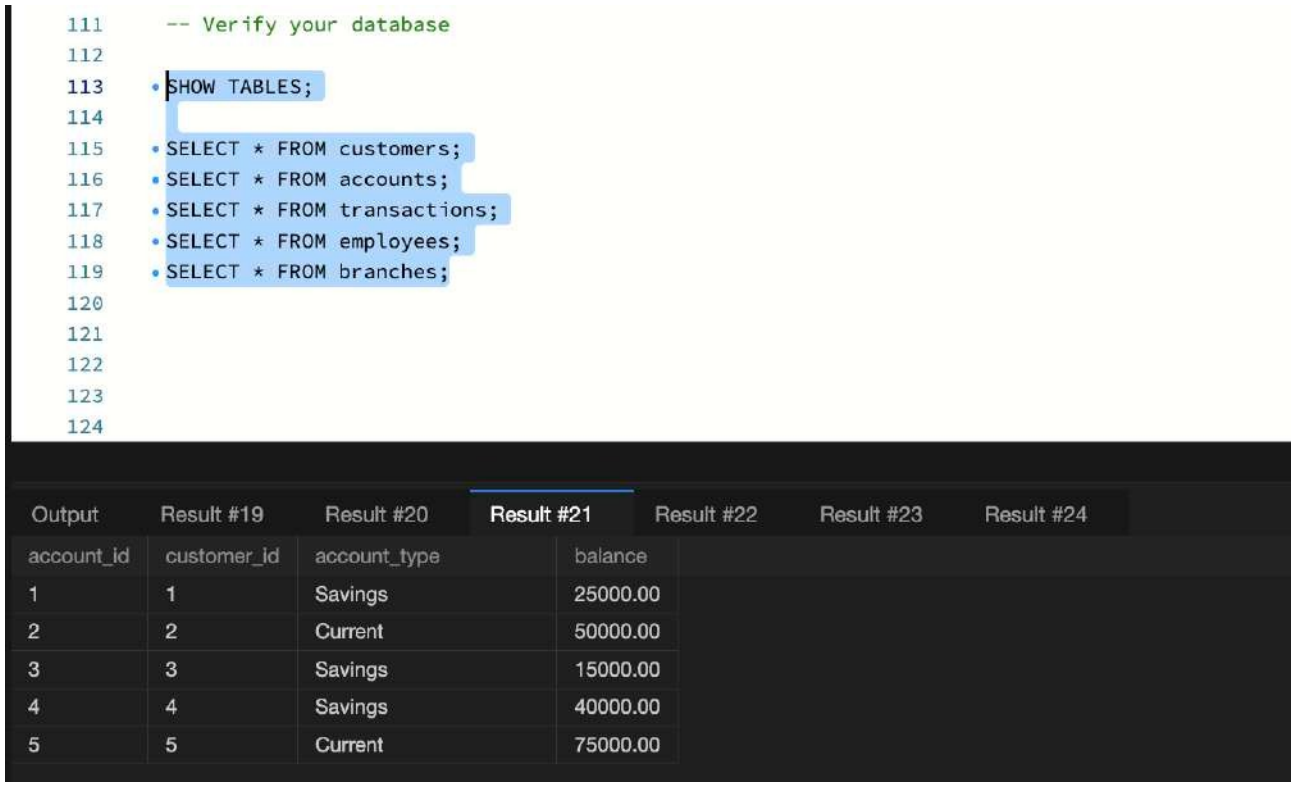
Output | Result #19 | Result #20 | Result #21 | Result #22 | Result #23 | Result #24

Tables_in_bank_management
accounts
branches
customers
employees
transactions

```
111    -- Verify your database
112
113    SHOW TABLES;
114
115    SELECT * FROM customers;
116    SELECT * FROM accounts;
117    SELECT * FROM transactions;
118    SELECT * FROM employees;
119    SELECT * FROM branches;
120
121
122
123
124
```

Output | Result #19 | **Result #20** | Result #21 | Result #22 | Result #23 | Result #24

customer_id	customer_name	email	city	phone
1	Rahul Sharma	rahul@gmail.com	Bangalore	9876543210
2	Priya Reddy	priya@gmail.com	Hyderabad	9876543211
3	Arjun Kumar	arjun@gmail.com	Chennai	9876543212
4	Sneha Rao	sneha@gmail.com	Bangalore	9876543213
5	Kiran Patel	kiran@gmail.com	Mumbai	9876543214



```
110
111     -- Verify your database
112
113     SHOW TABLES;
114
115     SELECT * FROM customers;
116     SELECT * FROM accounts;
117     SELECT * FROM transactions;
118     SELECT * FROM employees;
119     SELECT * FROM branches;
120
121
122
123
124
```

Output | Result #19 | Result #20 | Result #21 | **Result #22** | Result #23 | Result #24

transaction_id	account_id	transaction_type	amount	transaction_date
1	1	Deposit	5000.00	10/01/2026
2	2	Withdrawal	2000.00	10/02/2026
3	3	Deposit	3000.00	10/03/2026
4	4	Withdrawal	1500.00	10/04/2026
5	5	Deposit	10000.00	10/05/2026

```
112
113     SHOW TABLES;
114
115     SELECT * FROM customers;
116     SELECT * FROM accounts;
117     SELECT * FROM transactions;
118     SELECT * FROM employees;
119     SELECT * FROM branches;
120
121
122
123
124
```

Output | Result #19 | Result #20 | Result #21 | Result #22 | **Result #23** | Result #24

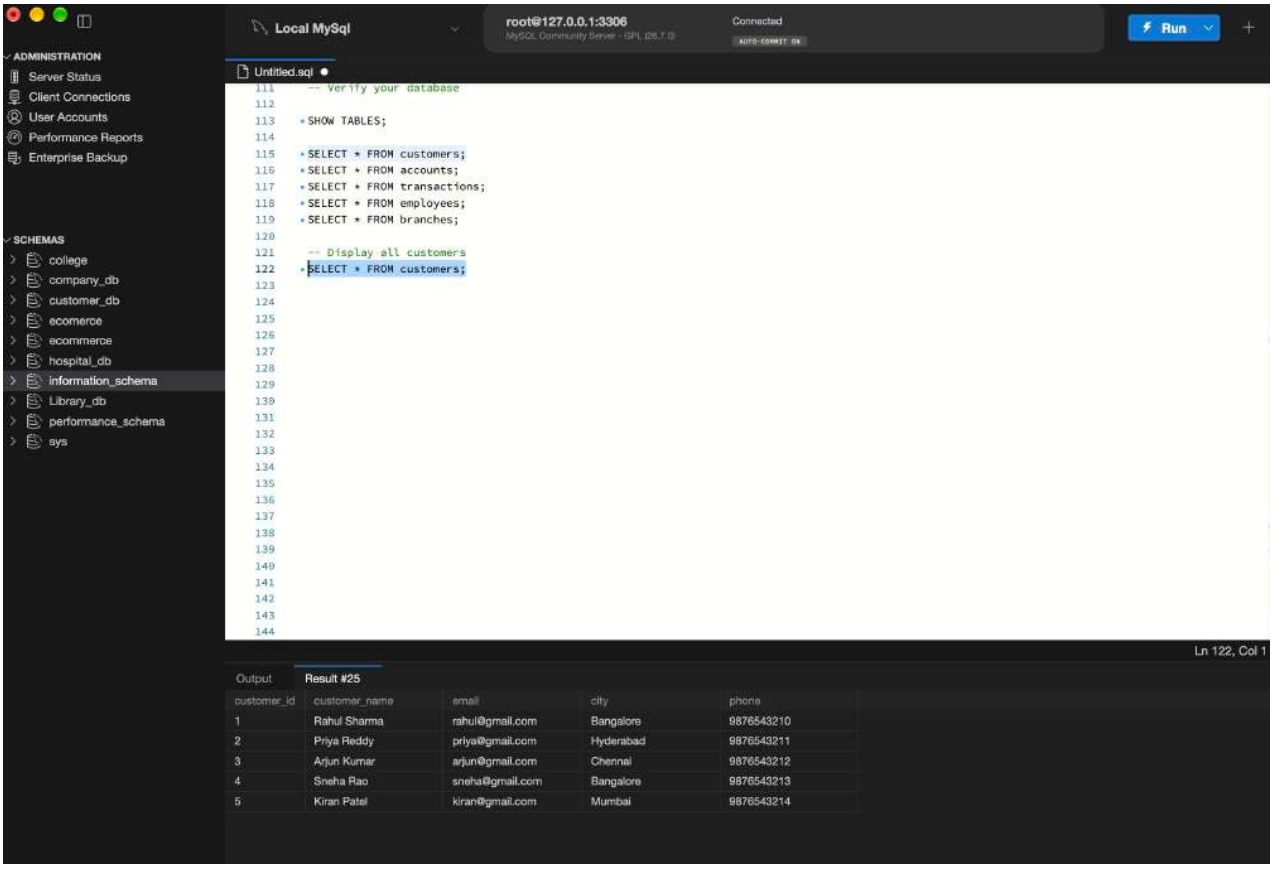
employee_id	employee_name	designation	salary	branch_city
1	Amit Verma	Manager	60000.00	Bangalore
2	Divya Singh	Cashier	35000.00	Hyderabad
3	Ravi Kumar	Accountant	45000.00	Chennai
4	Meena Das	Manager	65000.00	Mumbai

```
113  • SHOW TABLES;
114
115  • SELECT * FROM customers;
116  • SELECT * FROM accounts;
117  • SELECT * FROM transactions;
118  • SELECT * FROM employees;
119  • SELECT * FROM branches;
120
121
122
123
124
```

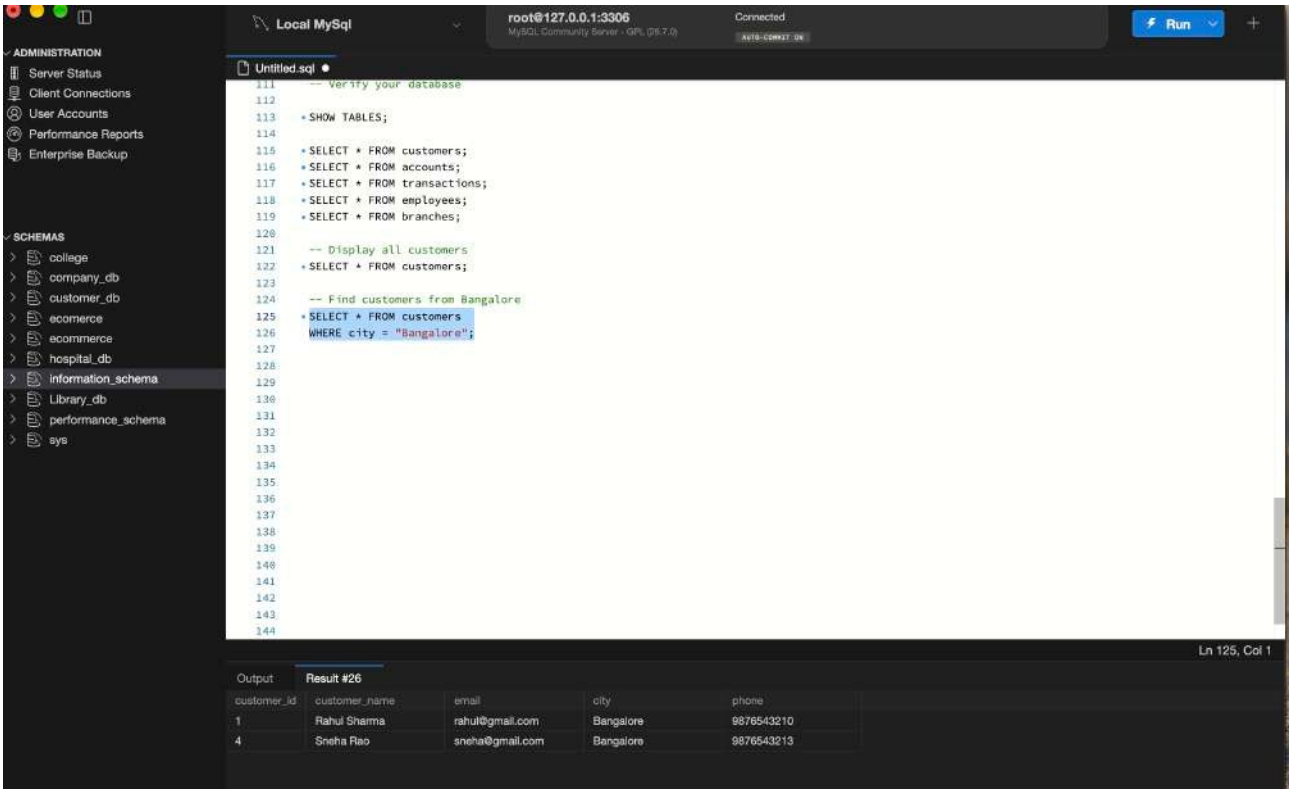
Output | Result #19 | Result #20 | Result #21 | Result #22 | Result #23 | **Result #24**

branch_id	branch_name	city	ifsc_code
1	Bangalore Main	Bangalore	BANK0001
2	Hyderabad Main	Hyderabad	BANK0002
3	Chennai Main	Chennai	BANK0003
4	Mumbai Main	Mumbai	BANK0004

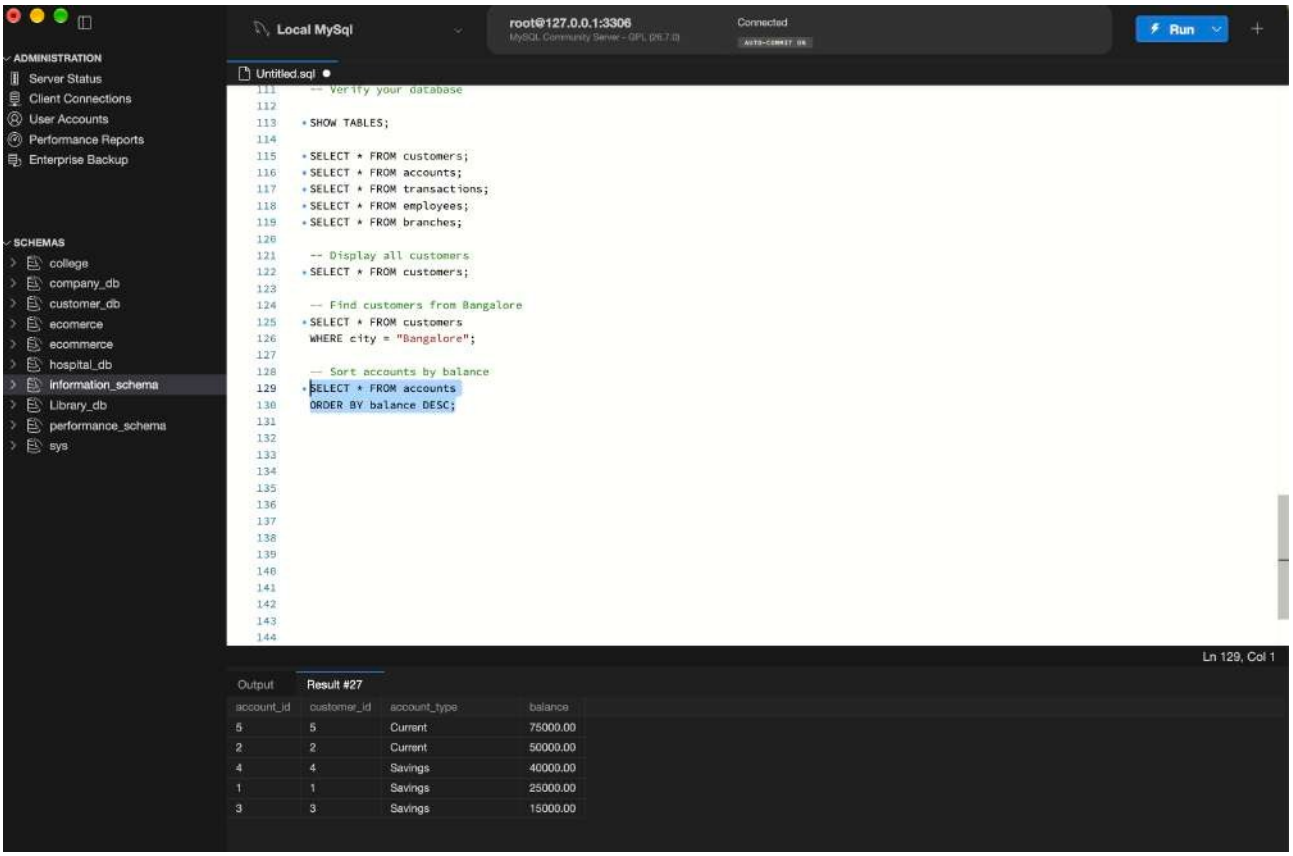
- **Display all customers**



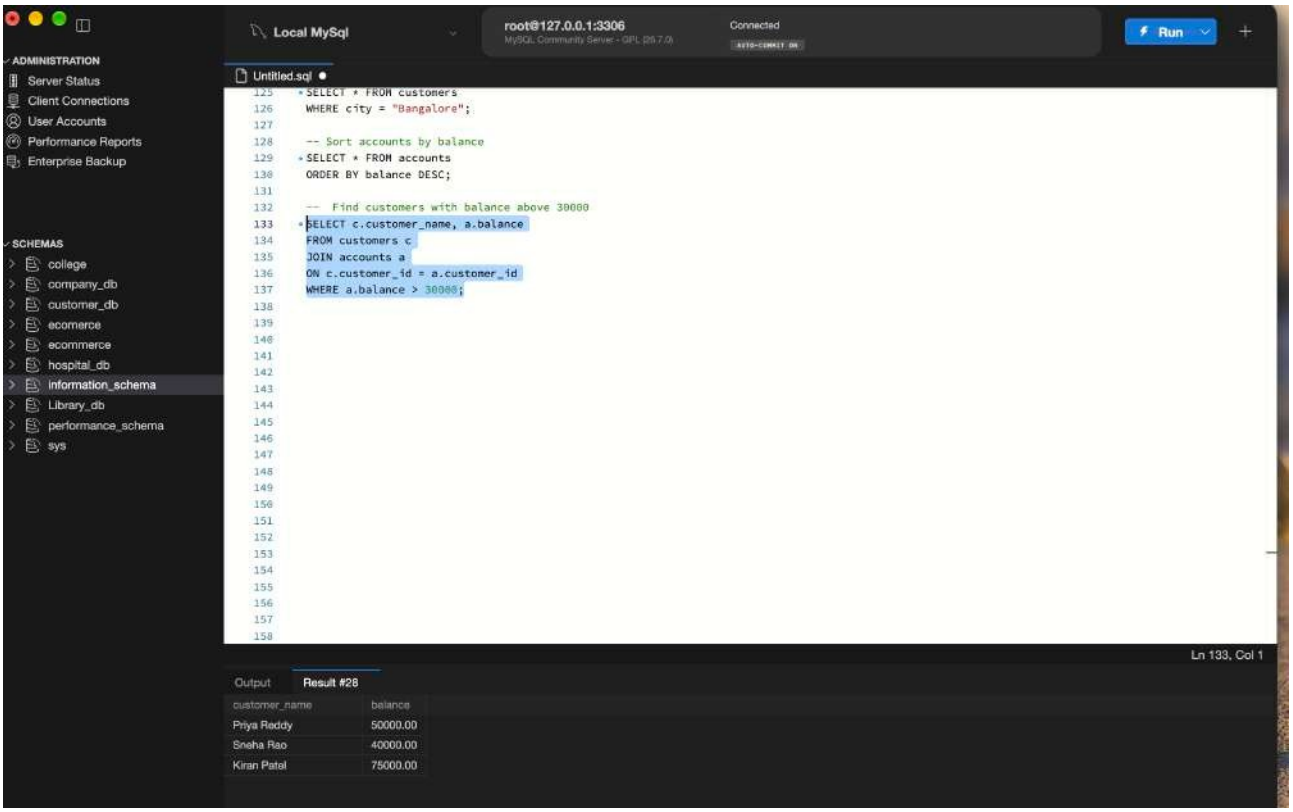
— Find customers from Bangalore



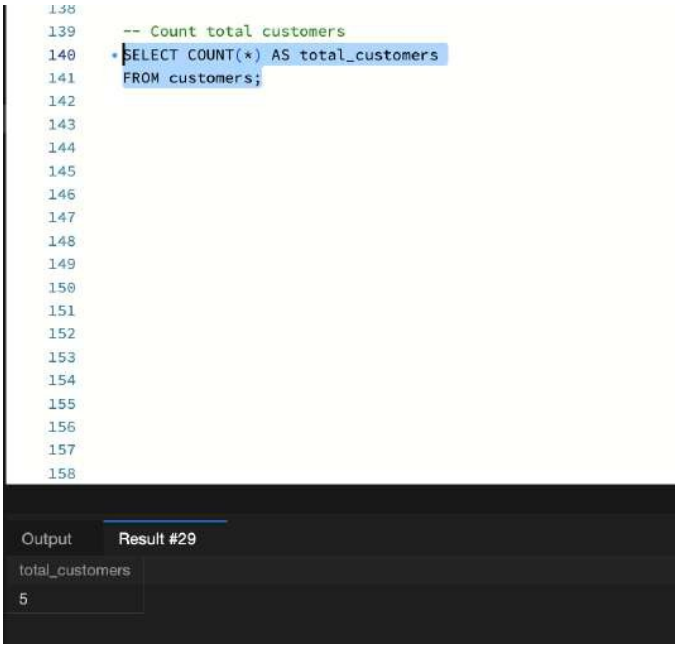
— Sort accounts by balance



— Find customers with balance above 30000



— Count total customers



— Calculate total bank balance

```
-- Calculate total bank balance
SELECT SUM(balance) AS total_balance
FROM accounts;
```

Ln 144, Col 1

Output | Result #30

total_balance
205000.00

— Calculate average account balance

```
-- Calculate average account balance
SELECT AVG(balance) AS average_balance
FROM accounts;
```

Ln 148, Col 1

Output | Result #31

average_balance
41000.000000

— Count accounts by account type

```
-- Count accounts by account type
SELECT account_type, COUNT(*) AS total_accounts
FROM accounts
GROUP BY account_type;
```

Ln 154, Col 23

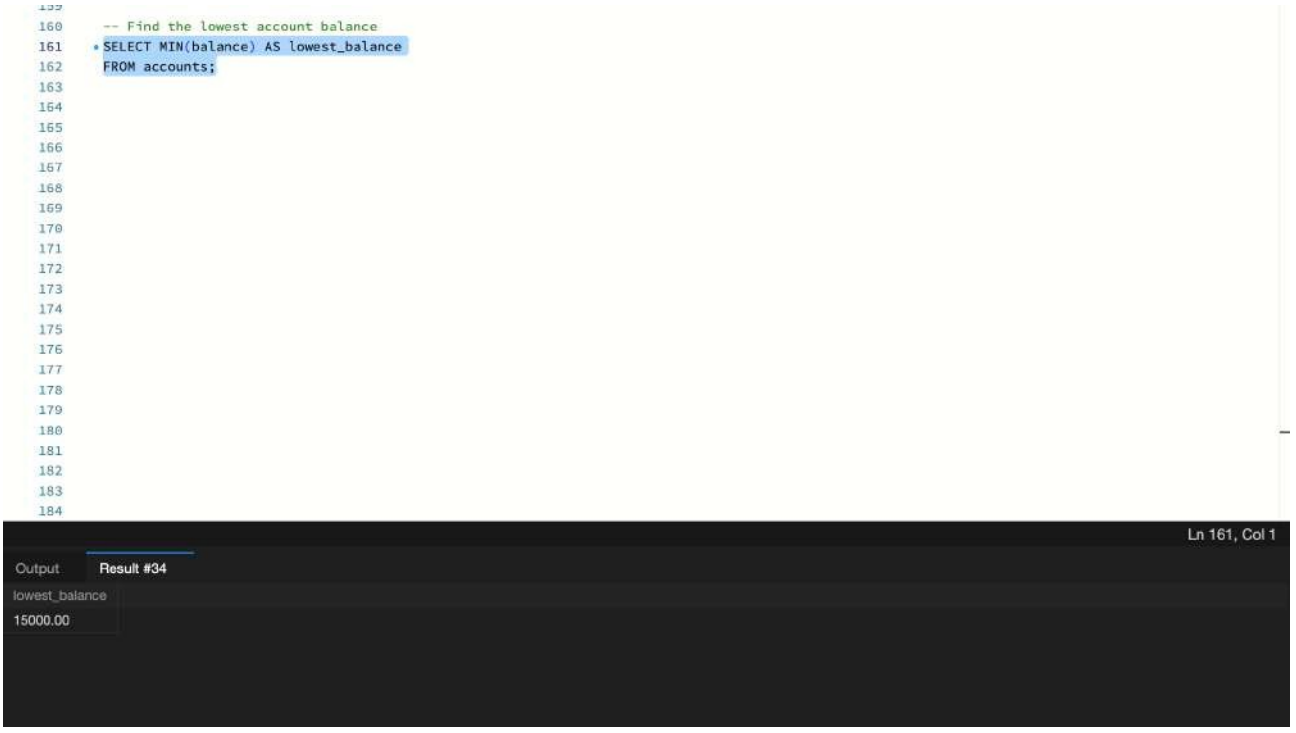
Output | Result #32

account_type	total_accounts
Savings	3
Current	2

— Find the highest account balance



— Find the lowest account balance



— Display transaction details with customer names

```sql
-- Display transaction details with customer names
SELECT c.customer_name, t.transaction_type,
       t.amount, t.transaction_date
FROM customers c
JOIN accounts a ON c.customer_id = a.customer_id
JOIN transactions t ON a.account_id = t.account_id;
```

Ln 166, Col 1

Output | Result #35

customer_name	transaction_type	amount	transaction_date
Rahul Sharma	Deposit	5000.00	10/01/2026
Priya Reddy	Withdrawal	2000.00	10/02/2026
Arjun Kumar	Deposit	3000.00	10/03/2026
Sneha Rao	Withdrawal	1500.00	10/04/2026
Kiran Patel	Deposit	10000.00	10/05/2026

-- Find deposit transactions

```sql
-- Find deposit transactions

SELECT * FROM transactions
WHERE transaction_type = 'Deposit';
```

Ln 174, Col 1

Output | Result #36

transaction_id	account_id	transaction_type	amount	transaction_date
1	1	Deposit	5000.00	10/01/2026
3	3	Deposit	3000.00	10/03/2026
5	5	Deposit	10000.00	10/05/2026

-- Find employees earning above 40000

```sql
-- Find employees earning above 40000

SELECT * FROM employees
WHERE salary > 40000;
```

Ln 178, Col 1

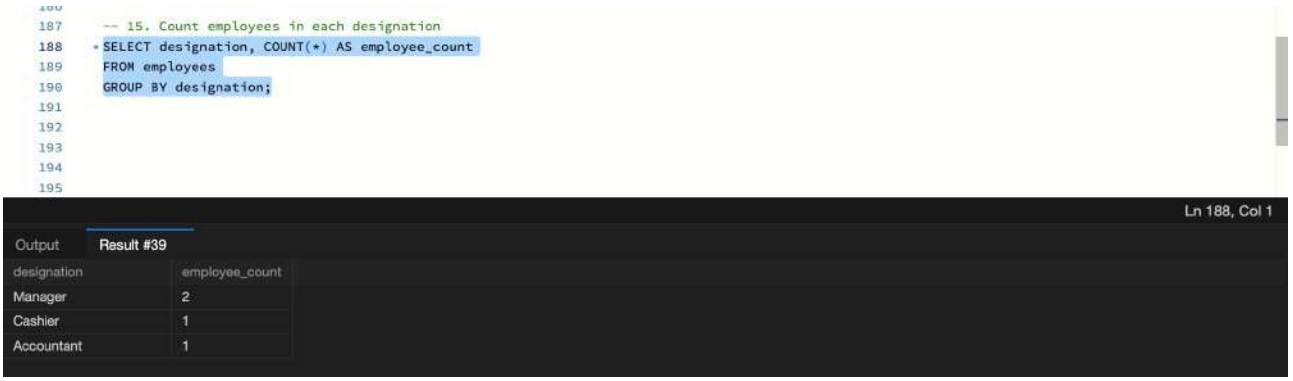
Output | Result #37

employee_id	employee_name	designation	salary	branch_city
1	Amit Verma	Manager	60000.00	Bangalore
3	Ravi Kumar	Accountant	45000.00	Chennai
4	Meena Das	Manager	65000.00	Mumbai

-- Calculate total salary expenditure



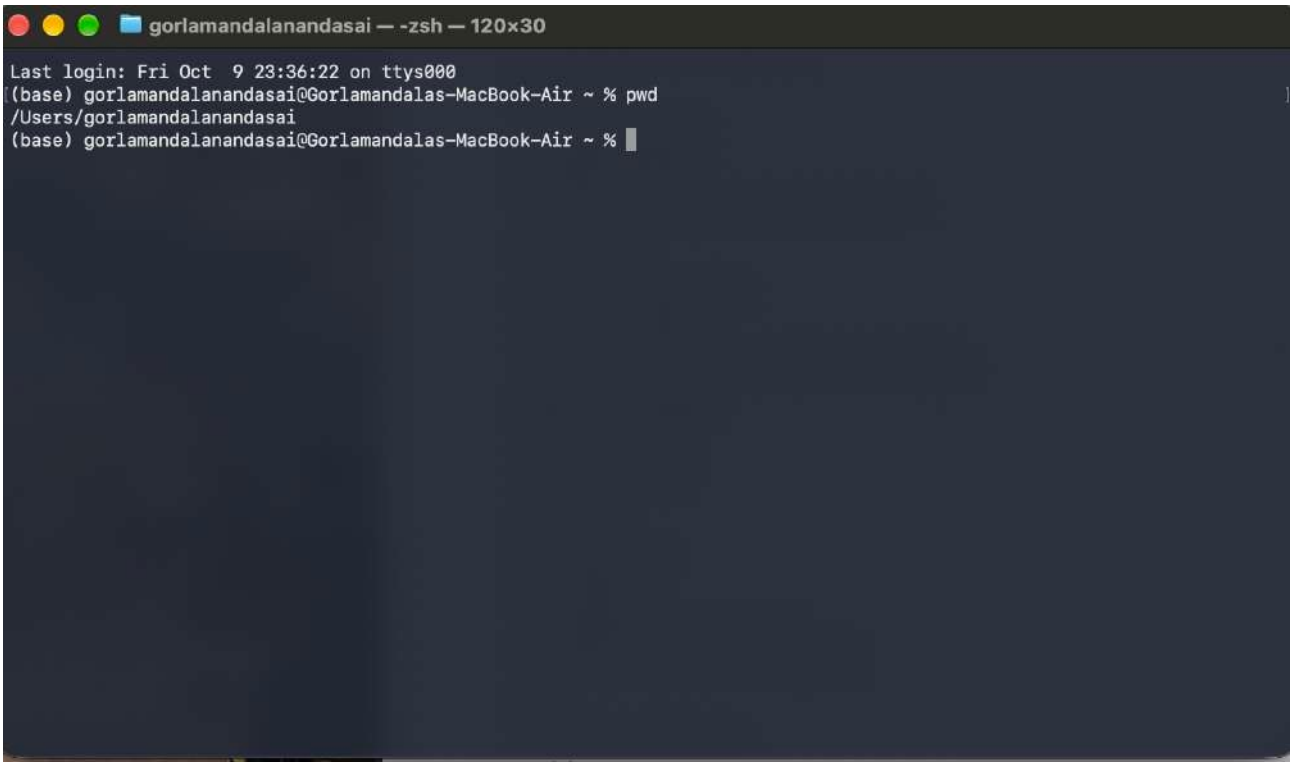
-- Count employees in each designation



Linux Practice

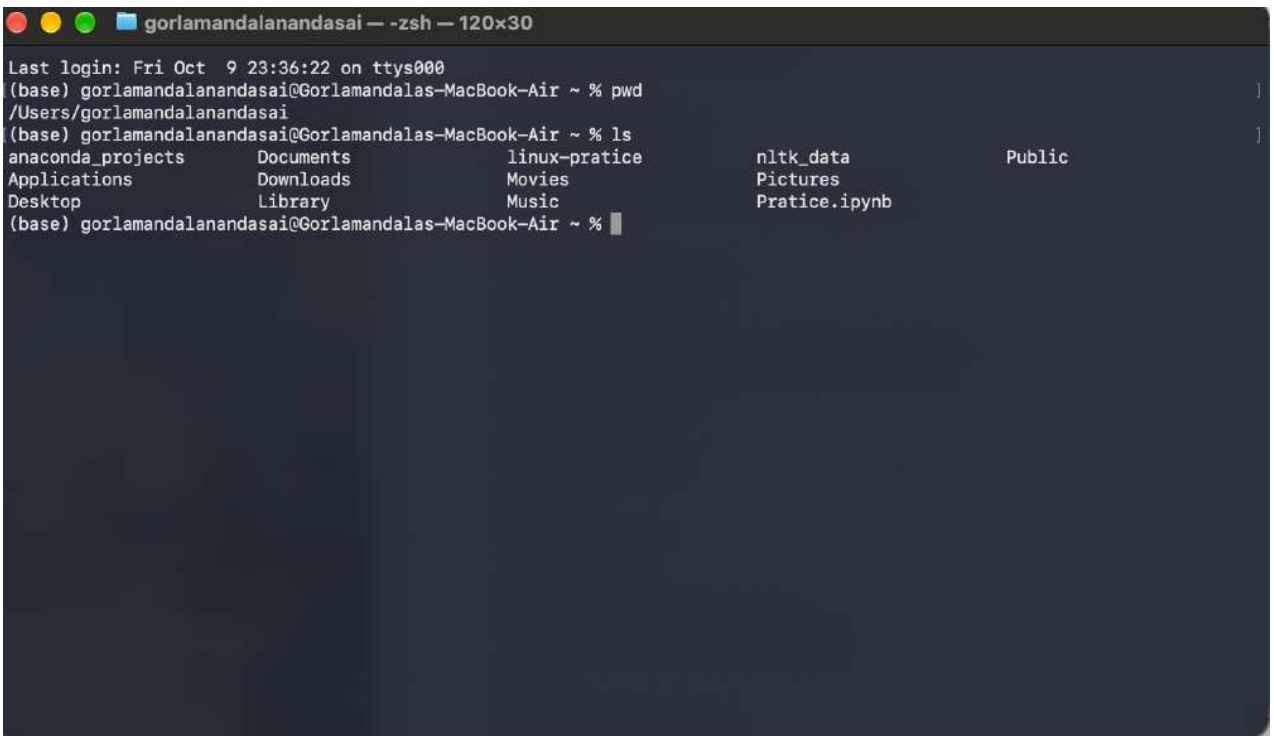
Command 1:

`pwd : Displays the current working directory`



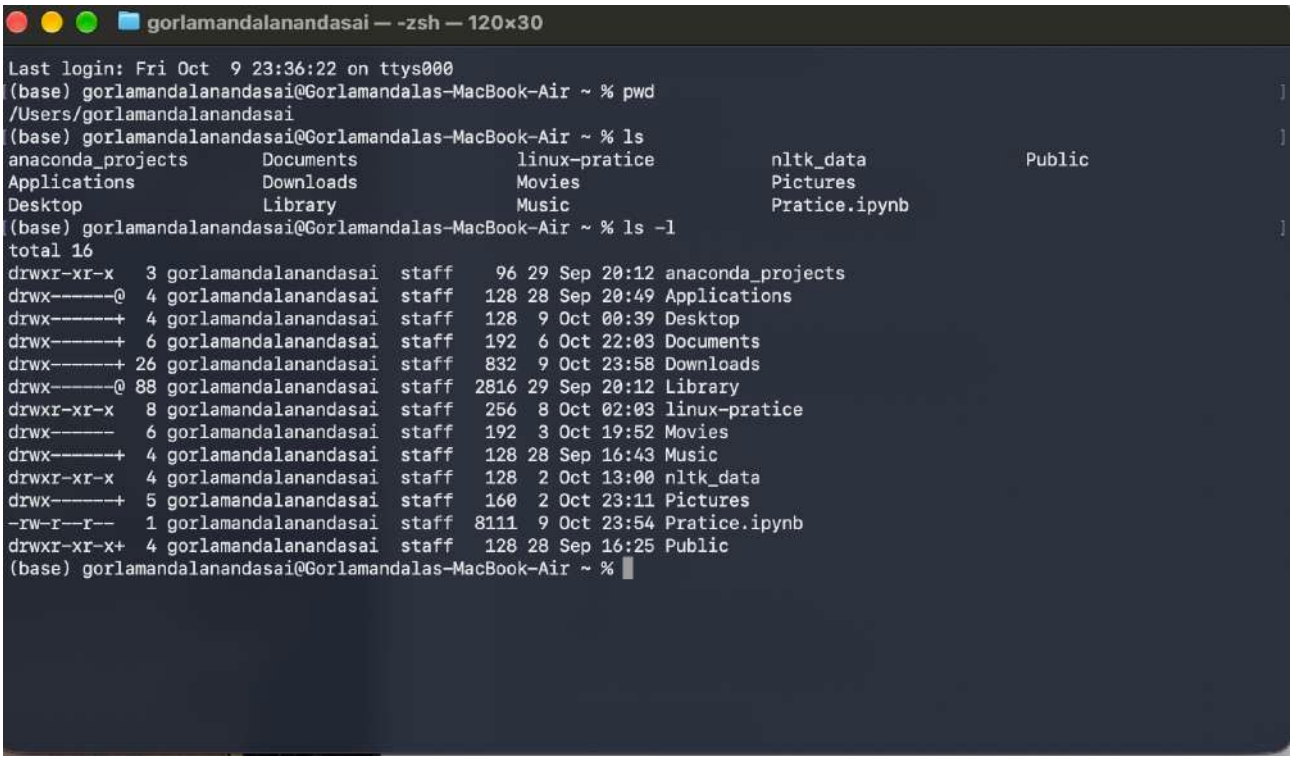
Command 2:

`ls : Lists the files and folders in your current directory.`



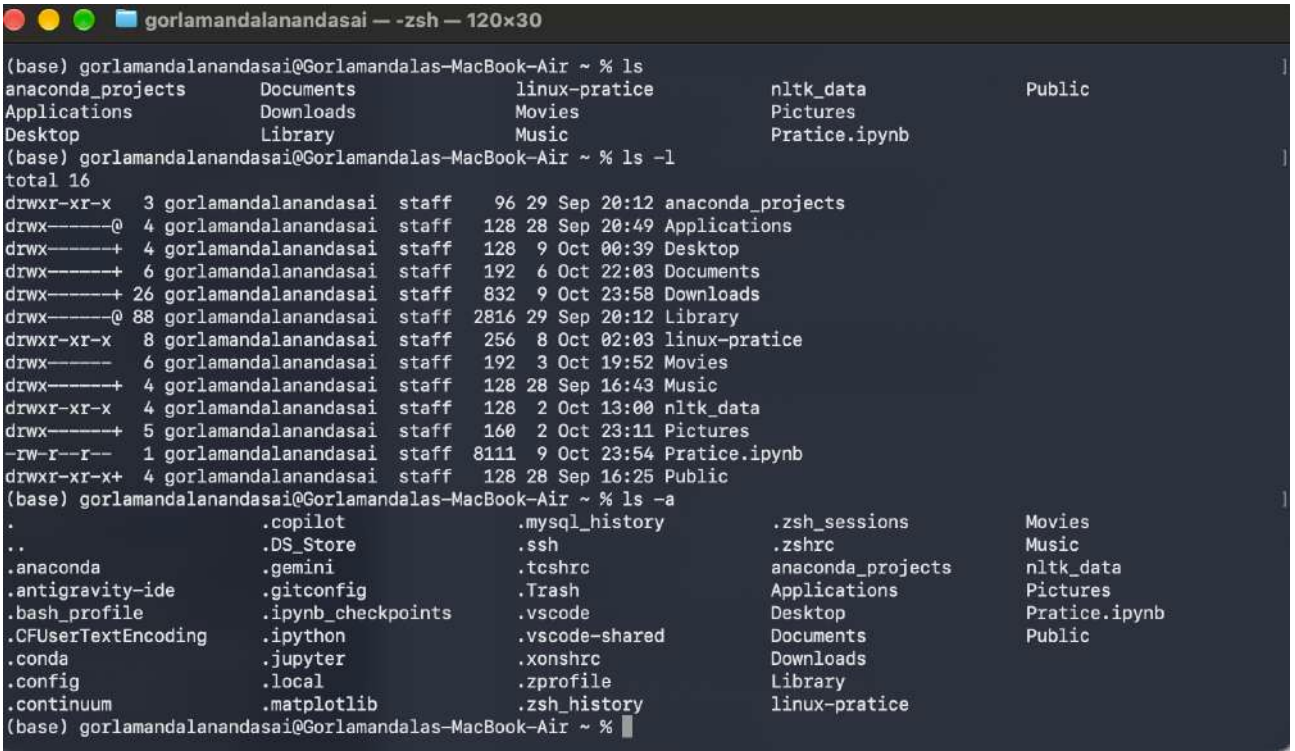
Command 3:

`ls -l` : Lists files and folders with detailed information, including permissions, owner, size, and modification date.



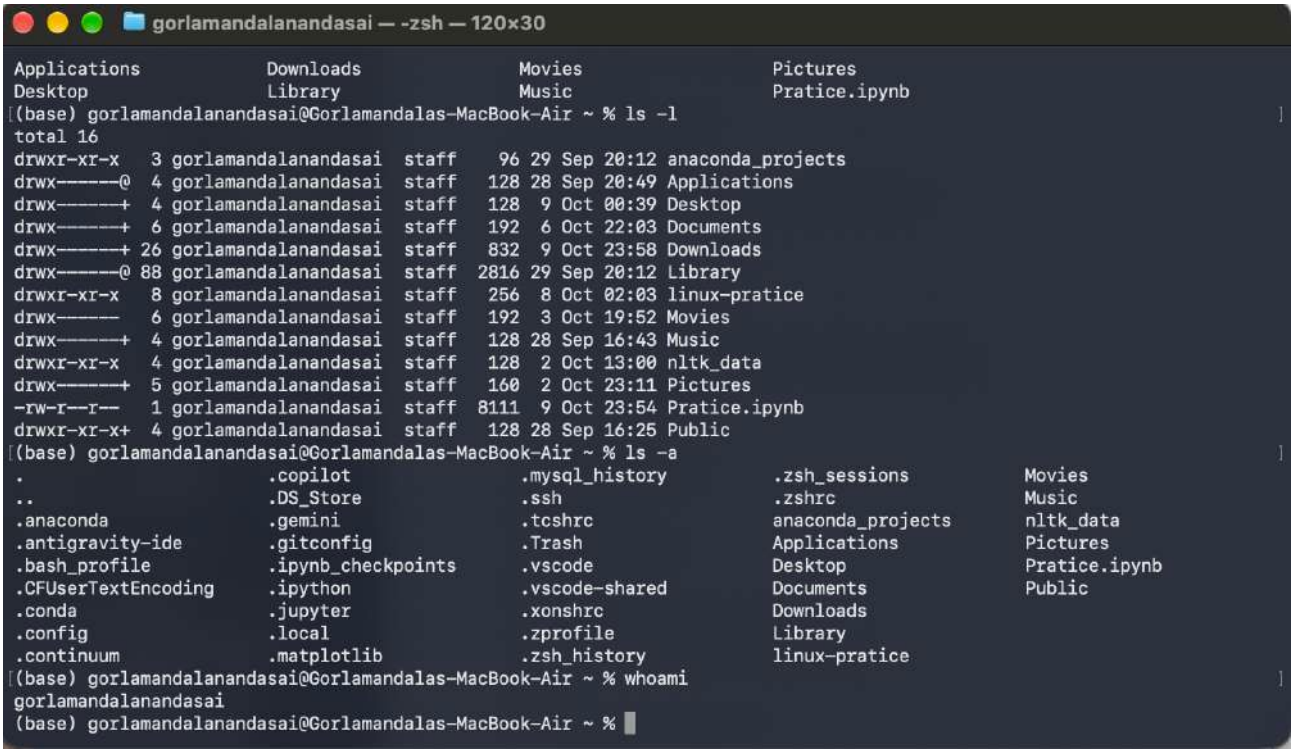
Command 4:

`ls -a` : Displays all files and folders, including hidden ones whose names begin with a dot (.)



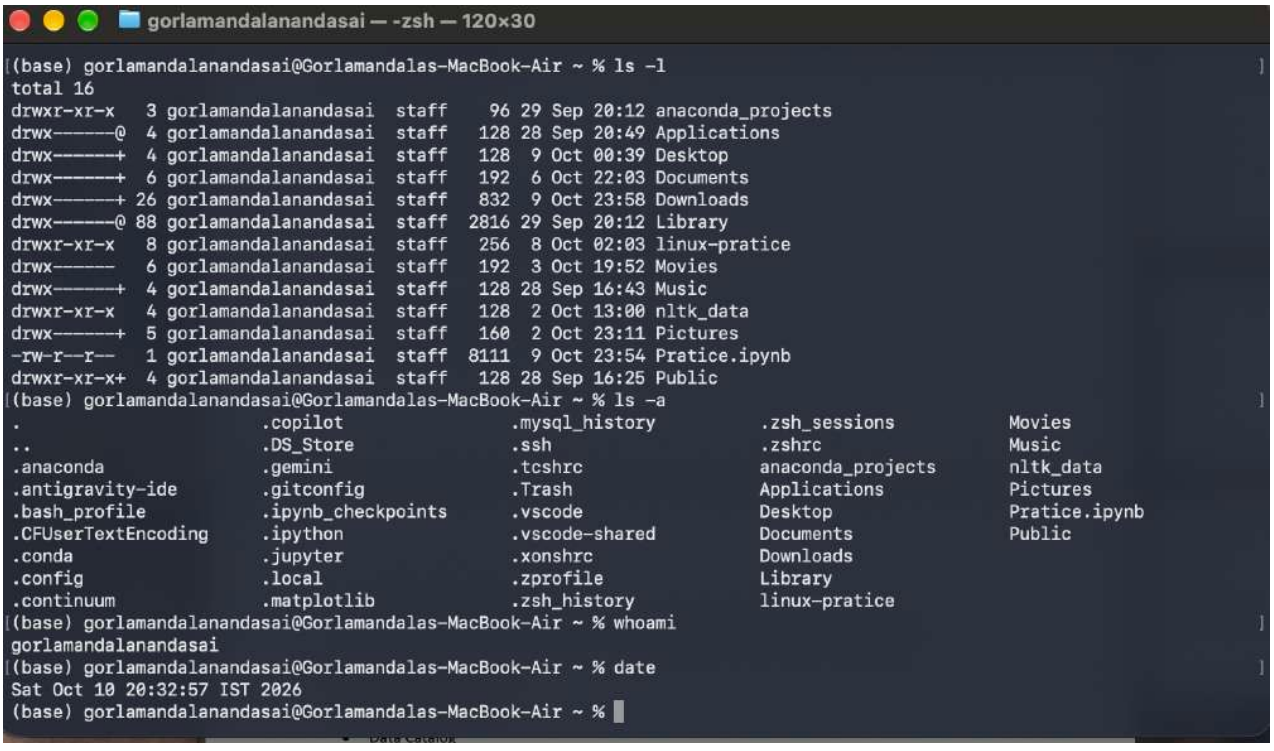
Command 5:

`whoami`: Displays the username of the current user.



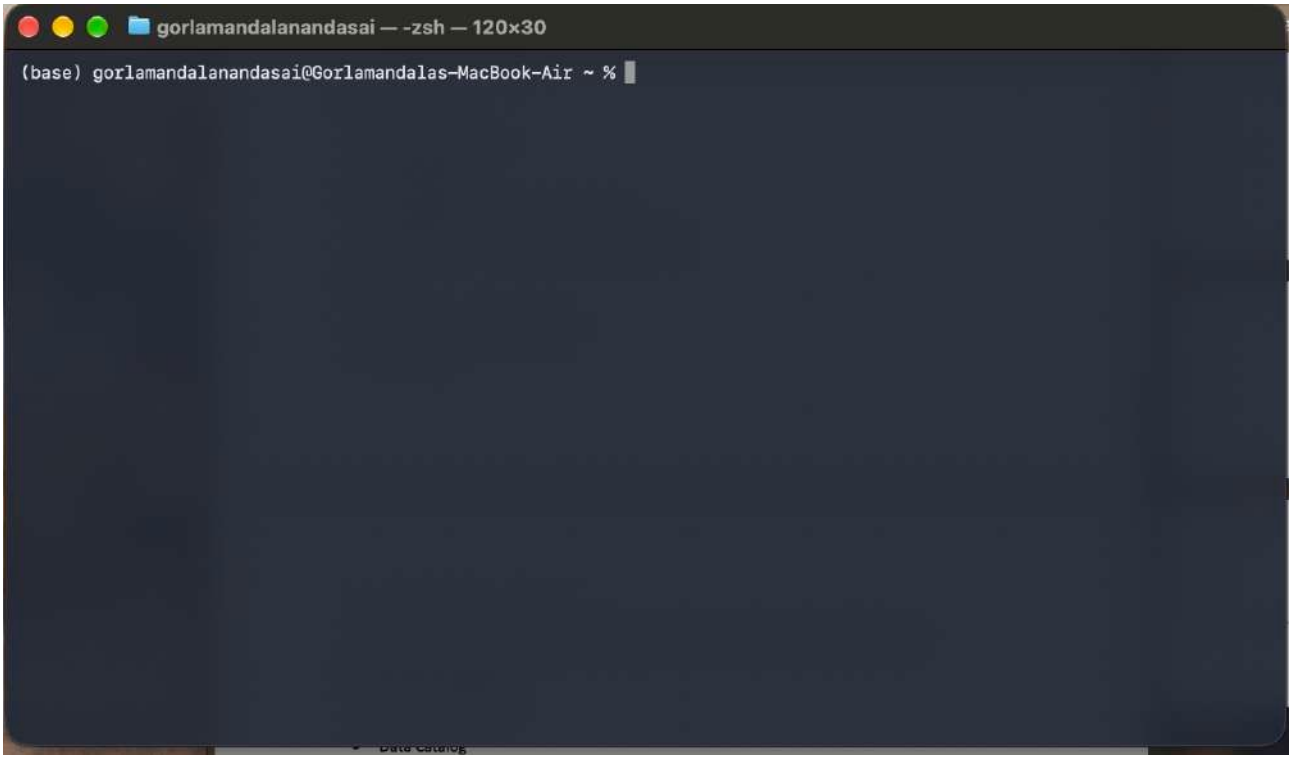
Command 6:

`date`: Displays the current date and time on your computer.

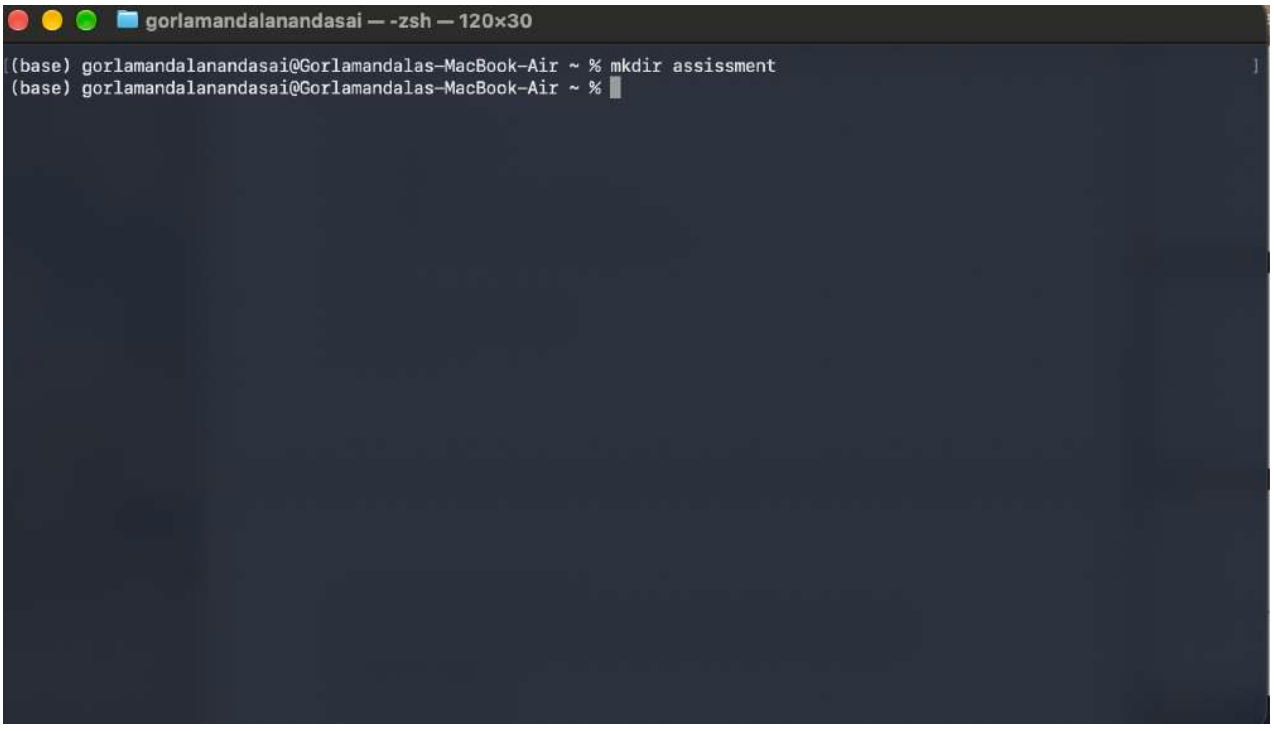


Command 7:

`clear`: Clears the visible terminal screen so you can work without old output cluttering it.

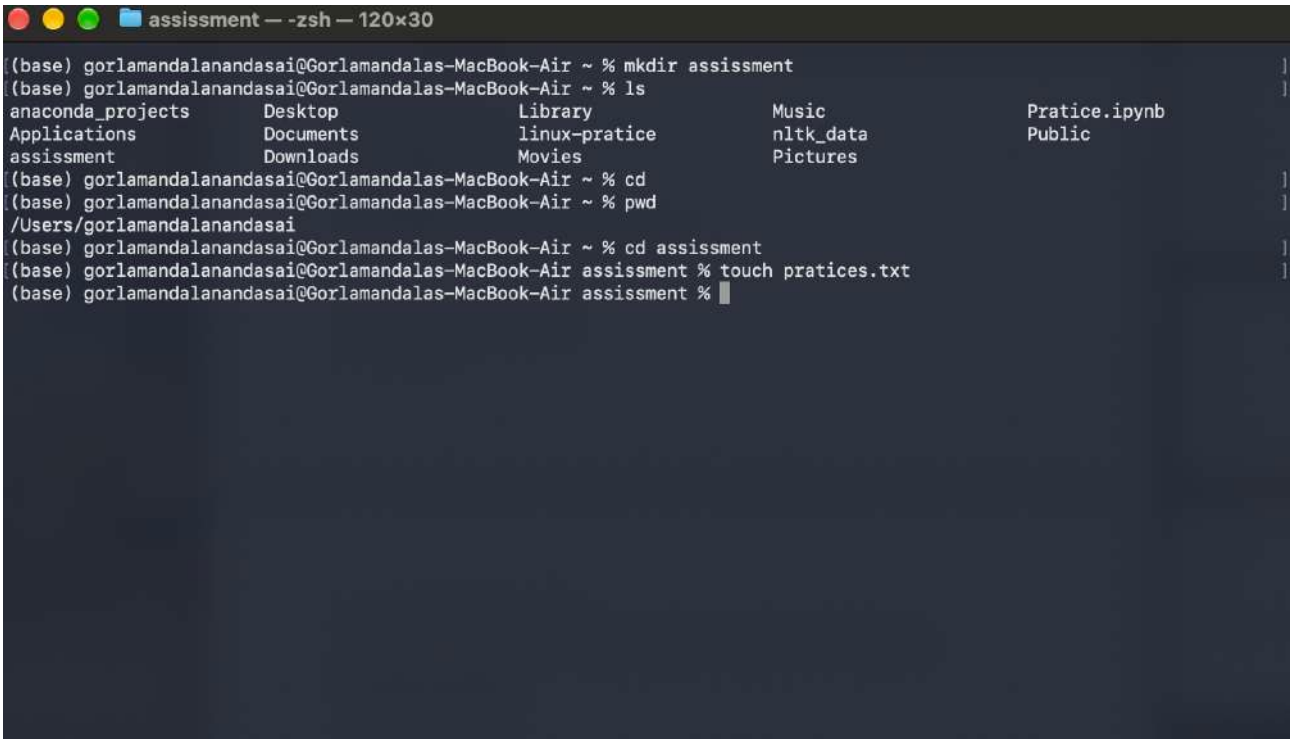


`mkdir: Creates a new directory (folder/file)`

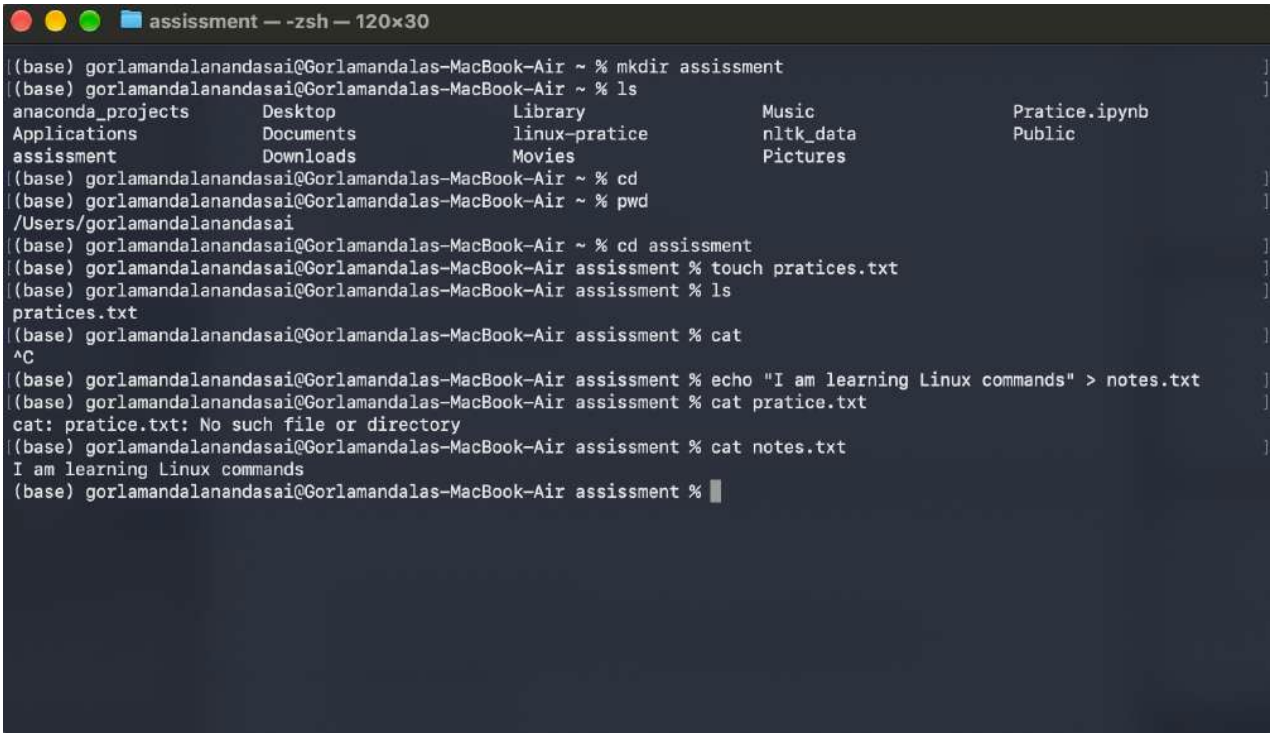


`touch`

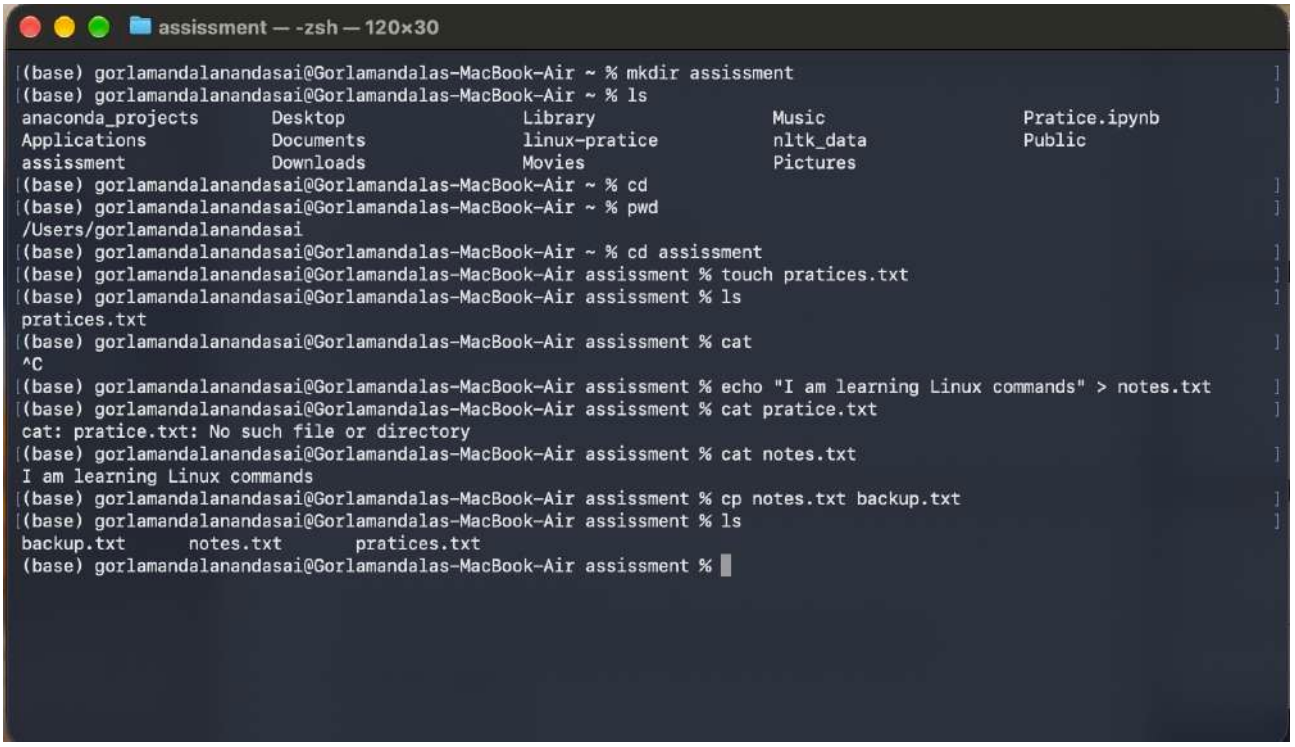
Creates a new empty file.

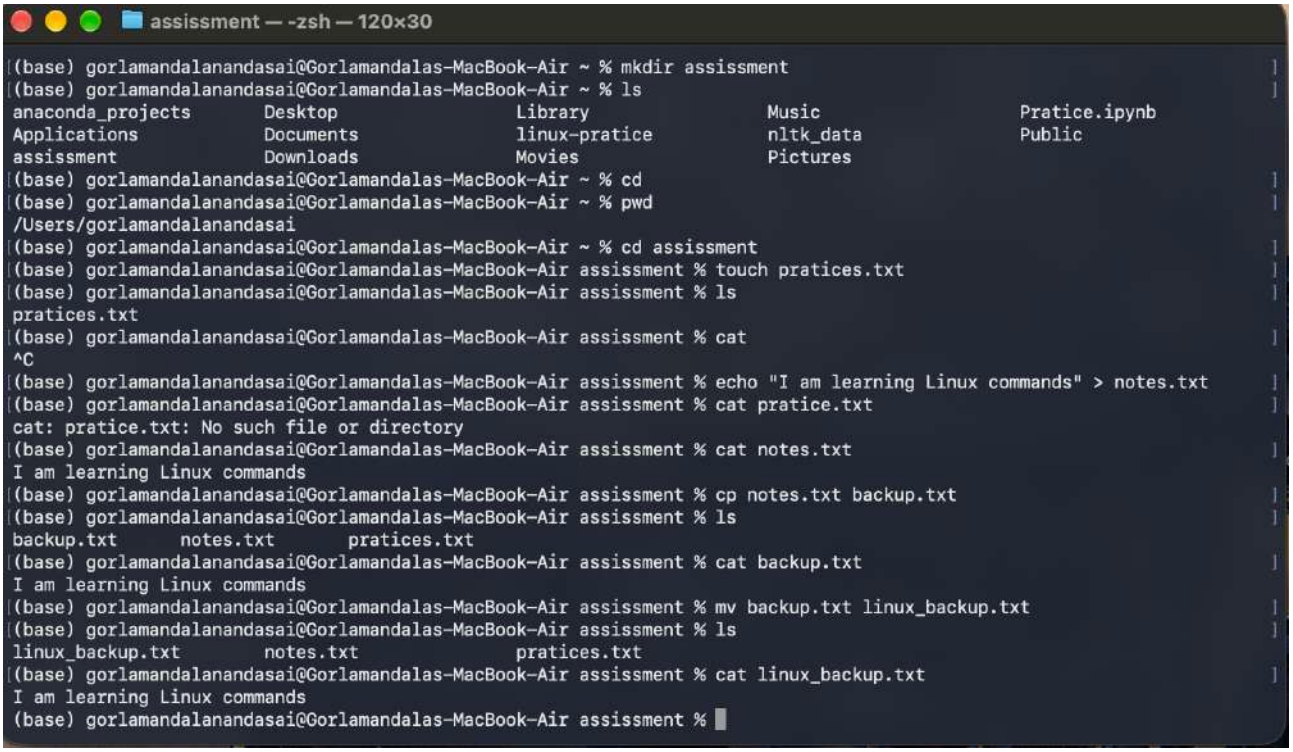


`cat` :Displays the contents of a text file in your terminal.

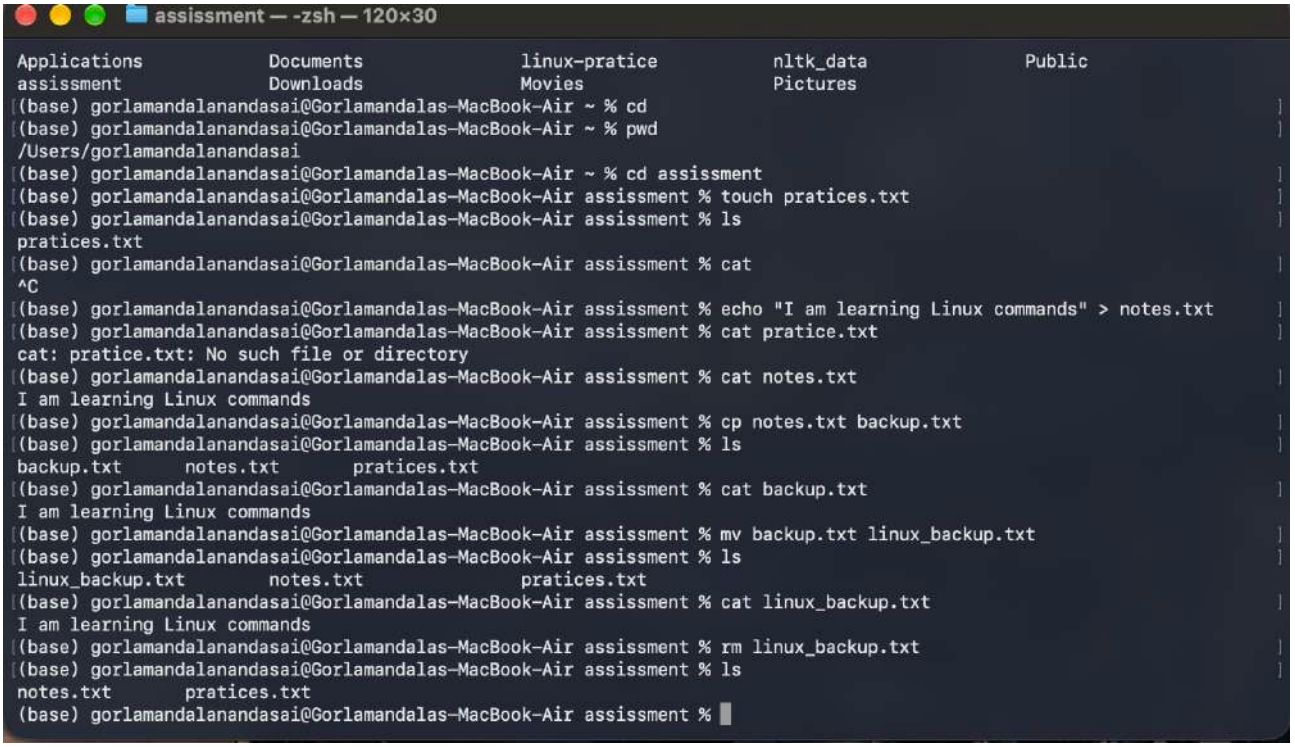


cp : Copies a file from one name to another

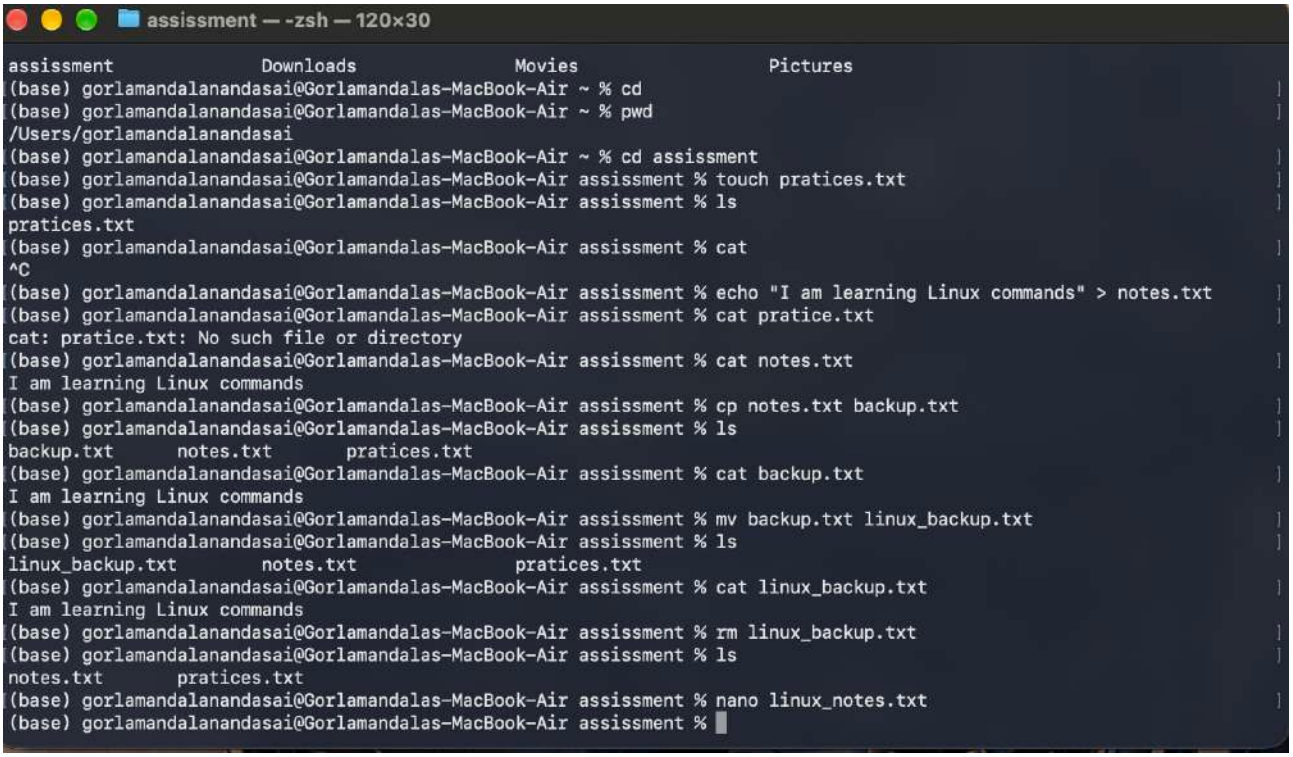


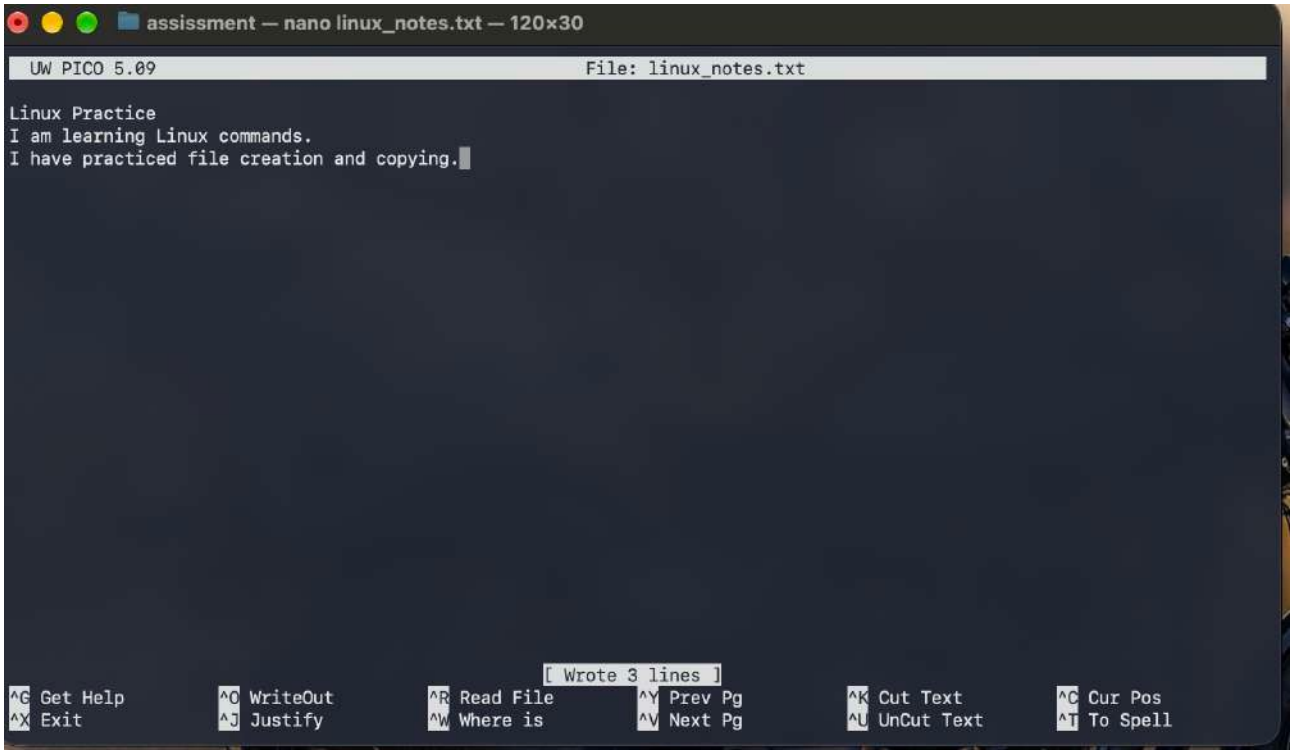


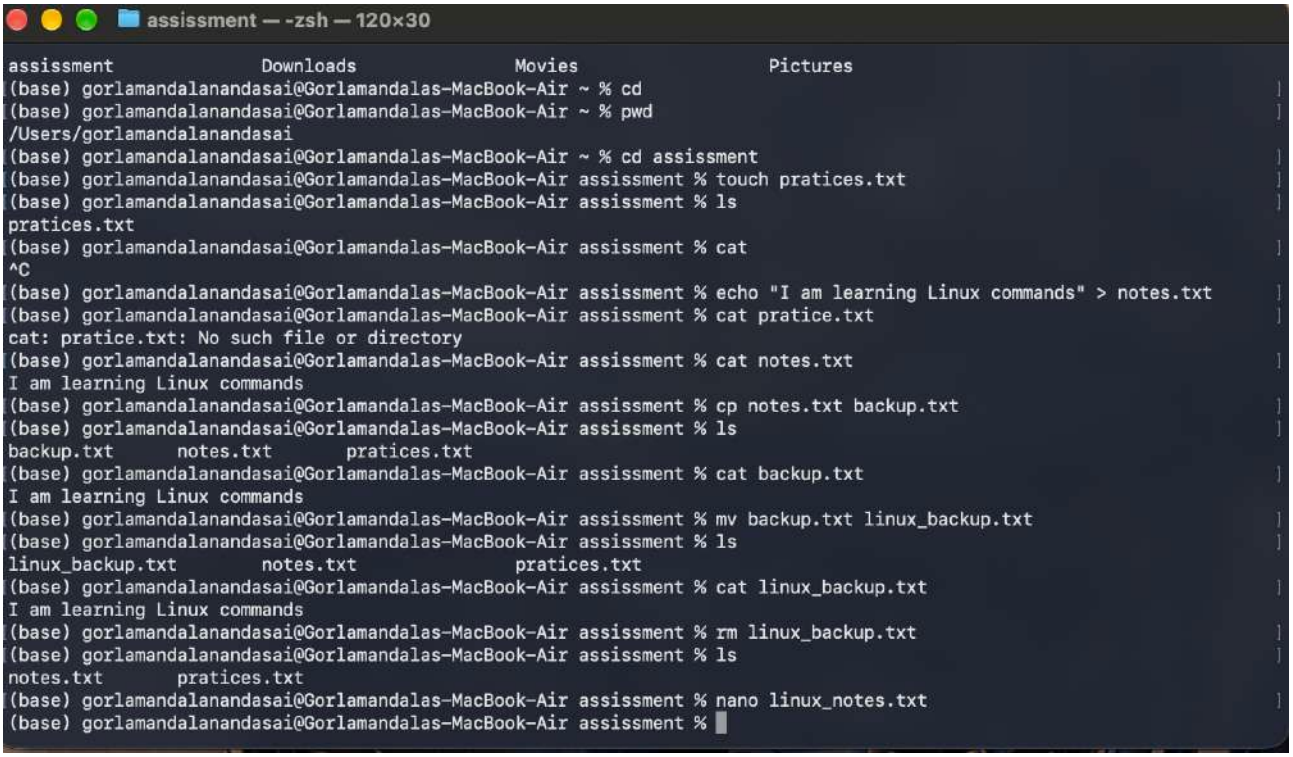
`rm` : Deletes a file from your current directory.

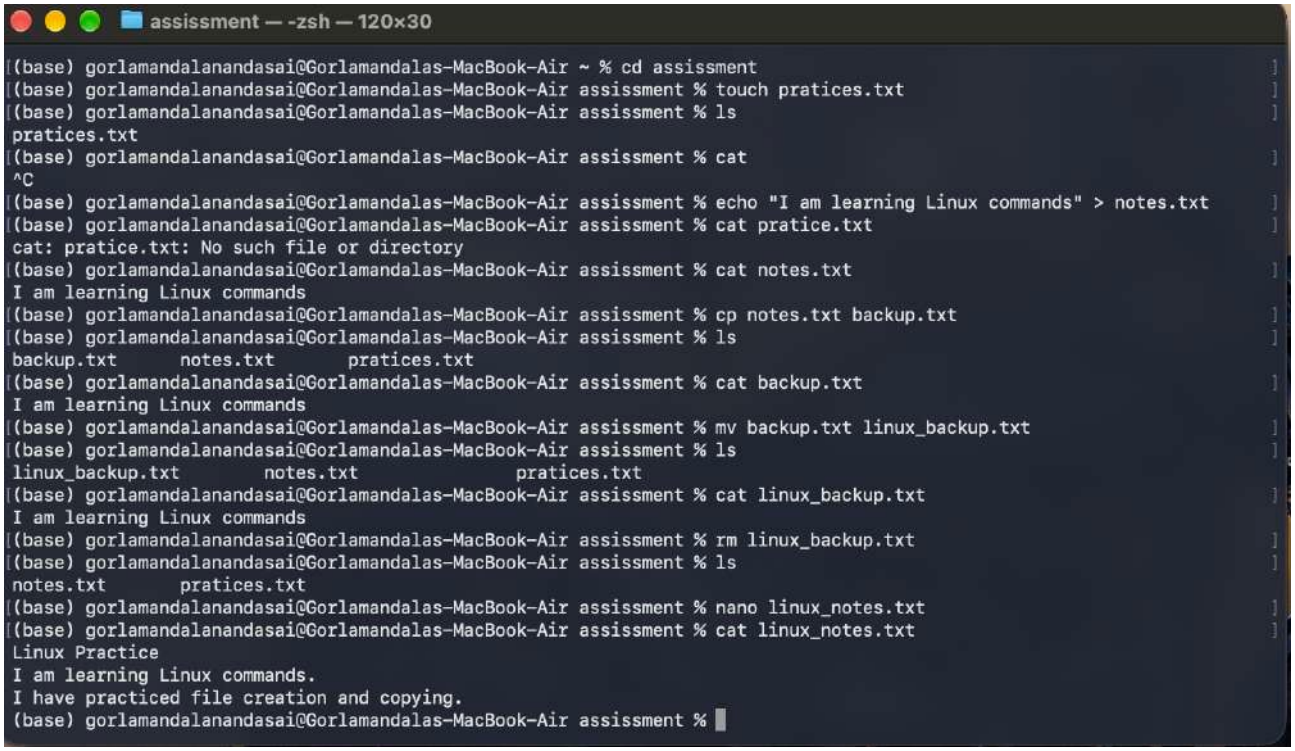


`nano`:Creates or edits a text file directly in the terminal.



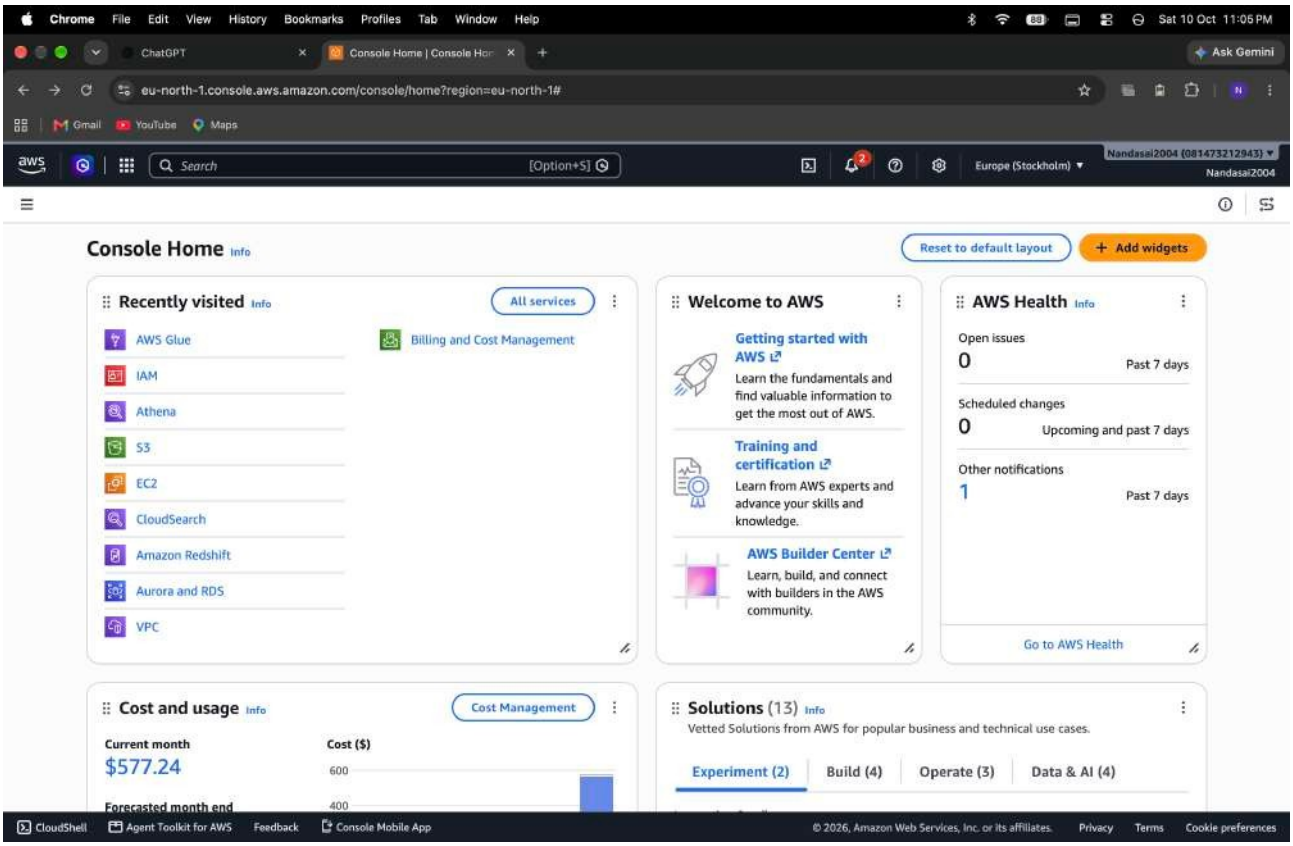

assissment — nano linux_notes.txt — 120×30
UW PICO 5.09
File: linux_notes.txt
Linux Practice
I am learning Linux commands.
I have practiced file creation and copying.
[Wrote 3 lines]
^G Get Help
^O WriteOut
^R Read File
^Y Prev Pg
^K Cut Text
^C Cur Pos
^X Exit
^J Justify
^W Where is
^V Next Pg
^U UnCut Text
^T To Spell


assissment — -zsh — 120×30
assissment Downloads Movies Pictures
(base) gorlamandalanandasai@Gorlamandalas-MacBook-Air ~ % cd
(base) gorlamandalanandasai@Gorlamandalas-MacBook-Air ~ % pwd
/Users/gorlamandalanandasai
(base) gorlamandalanandasai@Gorlamandalas-MacBook-Air ~ % cd assissment
(base) gorlamandalanandasai@Gorlamandalas-MacBook-Air assissment % touch pratices.txt
(base) gorlamandalanandasai@Gorlamandalas-MacBook-Air assissment % ls
pratices.txt
(base) gorlamandalanandasai@Gorlamandalas-MacBook-Air assissment % cat
^C
(base) gorlamandalanandasai@Gorlamandalas-MacBook-Air assissment % echo "I am learning Linux commands" > notes.txt
(base) gorlamandalanandasai@Gorlamandalas-MacBook-Air assissment % cat pratice.txt
cat: pratice.txt: No such file or directory
(base) gorlamandalanandasai@Gorlamandalas-MacBook-Air assissment % cat notes.txt
I am learning Linux commands
(base) gorlamandalanandasai@Gorlamandalas-MacBook-Air assissment % cp notes.txt backup.txt
(base) gorlamandalanandasai@Gorlamandalas-MacBook-Air assissment % ls
backup.txt notes.txt pratices.txt
(base) gorlamandalanandasai@Gorlamandalas-MacBook-Air assissment % cat backup.txt
I am learning Linux commands
(base) gorlamandalanandasai@Gorlamandalas-MacBook-Air assissment % mv backup.txt linux_backup.txt
(base) gorlamandalanandasai@Gorlamandalas-MacBook-Air assissment % ls
linux_backup.txt notes.txt pratices.txt
(base) gorlamandalanandasai@Gorlamandalas-MacBook-Air assissment % cat linux_backup.txt
I am learning Linux commands
(base) gorlamandalanandasai@Gorlamandalas-MacBook-Air assissment % rm linux_backup.txt
(base) gorlamandalanandasai@Gorlamandalas-MacBook-Air assissment % ls
notes.txt pratices.txt
(base) gorlamandalanandasai@Gorlamandalas-MacBook-Air assissment % nano linux_notes.txt
(base) gorlamandalanandasai@Gorlamandalas-MacBook-Air assissment %

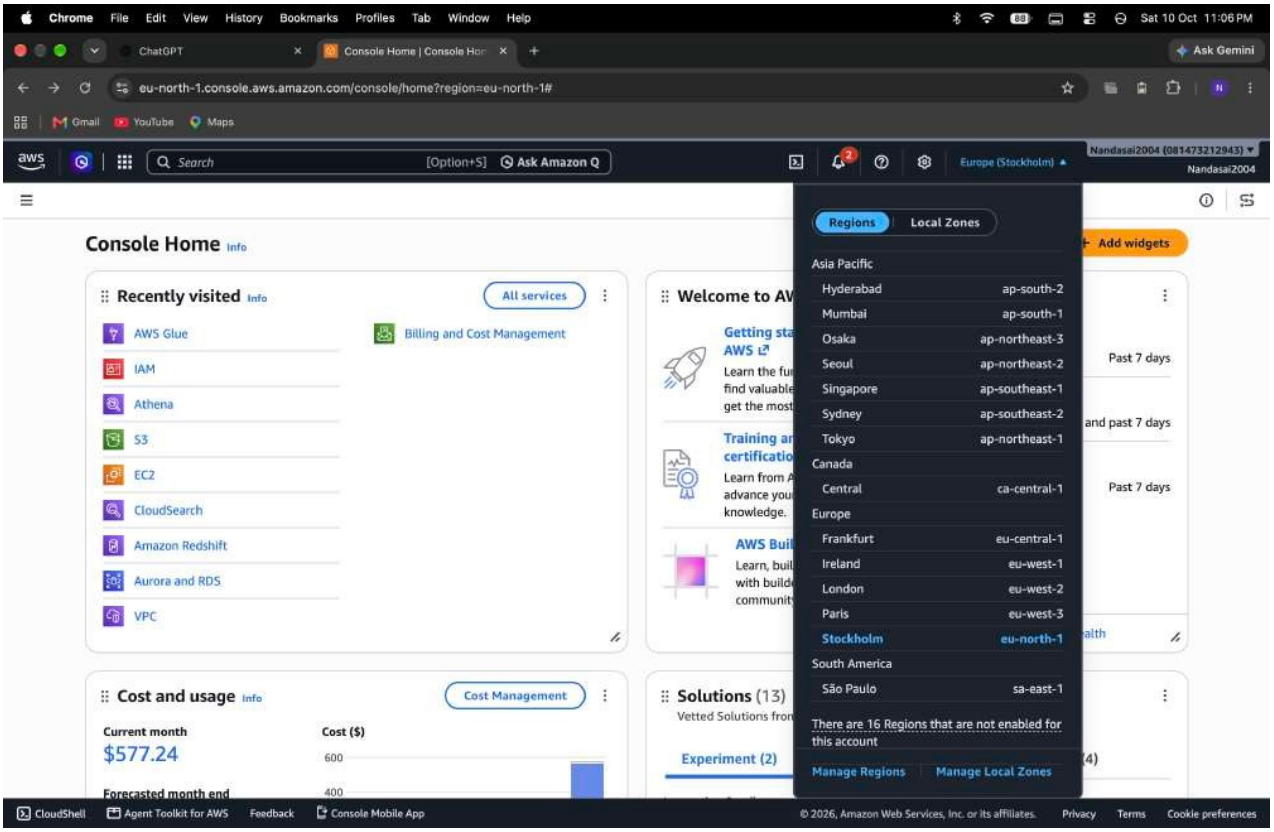

assissment — -zsh — 120×30
(base) gorlamandalanandasai@Gorlamandalas-MacBook-Air ~ % cd assissment
(base) gorlamandalanandasai@Gorlamandalas-MacBook-Air assissment % touch pratices.txt
(base) gorlamandalanandasai@Gorlamandalas-MacBook-Air assissment % ls
pratices.txt
(base) gorlamandalanandasai@Gorlamandalas-MacBook-Air assissment % cat
^C
(base) gorlamandalanandasai@Gorlamandalas-MacBook-Air assissment % echo "I am learning Linux commands" > notes.txt
(base) gorlamandalanandasai@Gorlamandalas-MacBook-Air assissment % cat pratice.txt
cat: pratice.txt: No such file or directory
(base) gorlamandalanandasai@Gorlamandalas-MacBook-Air assissment % cat notes.txt
I am learning Linux commands
(base) gorlamandalanandasai@Gorlamandalas-MacBook-Air assissment % cp notes.txt backup.txt
(base) gorlamandalanandasai@Gorlamandalas-MacBook-Air assissment % ls
backup.txt notes.txt pratices.txt
(base) gorlamandalanandasai@Gorlamandalas-MacBook-Air assissment % cat backup.txt
I am learning Linux commands
(base) gorlamandalanandasai@Gorlamandalas-MacBook-Air assissment % mv backup.txt linux_backup.txt
(base) gorlamandalanandasai@Gorlamandalas-MacBook-Air assissment % ls
linux_backup.txt notes.txt pratices.txt
(base) gorlamandalanandasai@Gorlamandalas-MacBook-Air assissment % cat linux_backup.txt
I am learning Linux commands
(base) gorlamandalanandasai@Gorlamandalas-MacBook-Air assissment % rm linux_backup.txt
(base) gorlamandalanandasai@Gorlamandalas-MacBook-Air assissment % ls
notes.txt pratices.txt
(base) gorlamandalanandasai@Gorlamandalas-MacBook-Air assissment % nano linux_notes.txt
(base) gorlamandalanandasai@Gorlamandalas-MacBook-Air assissment % cat linux_notes.txt
Linux Practice
I am learning Linux commands.
I have practiced file creation and copying.
(base) gorlamandalanandasai@Gorlamandalas-MacBook-Air assissment %

AWS Account / Region

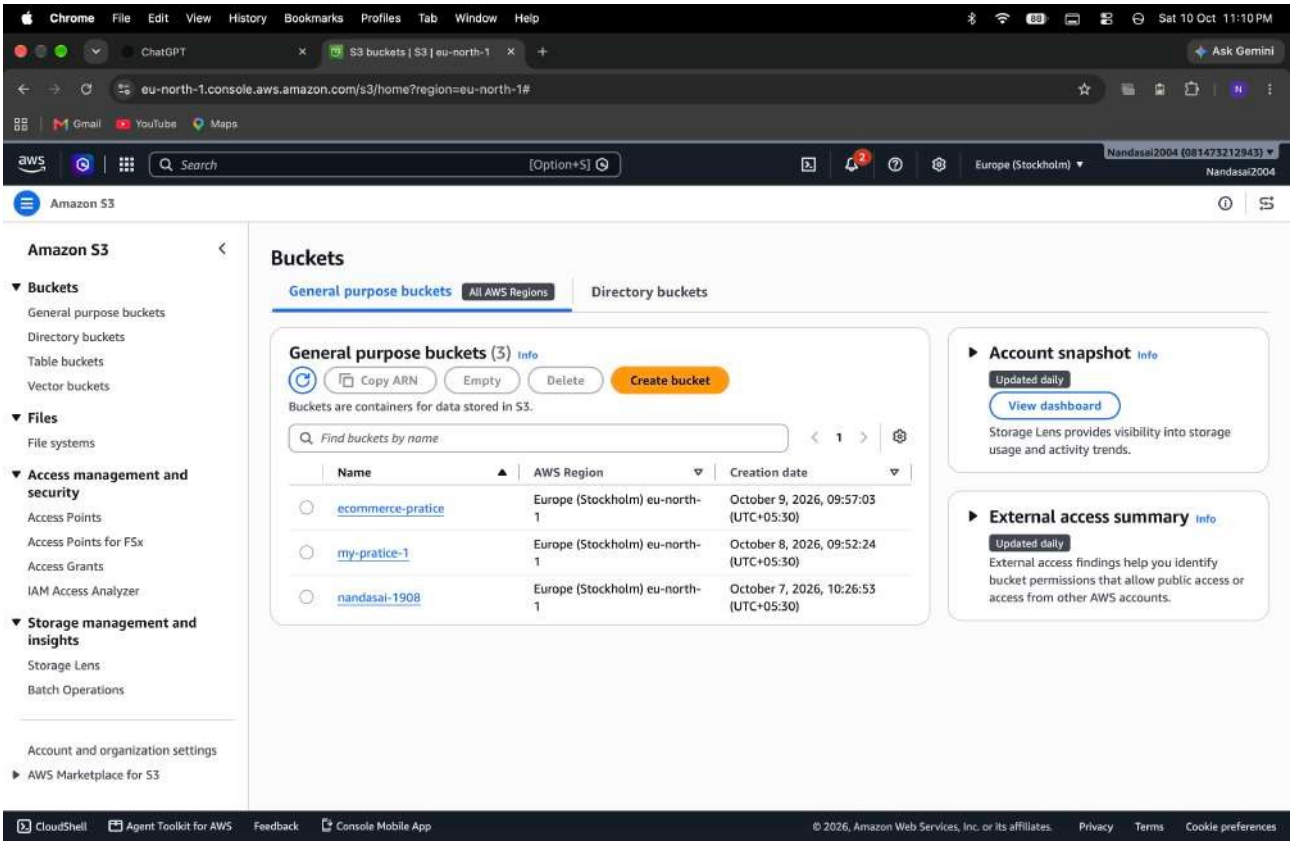
AWS Console:

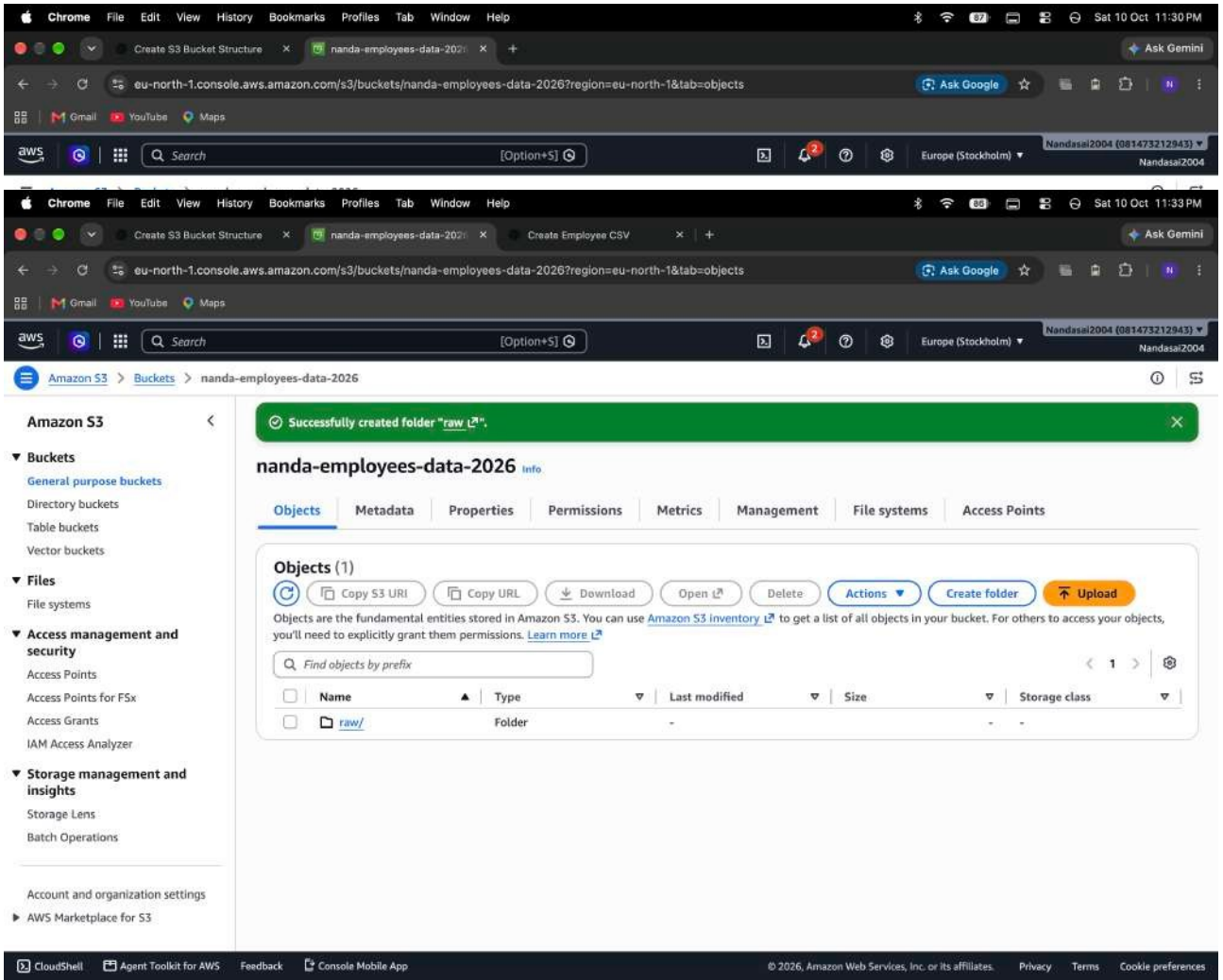


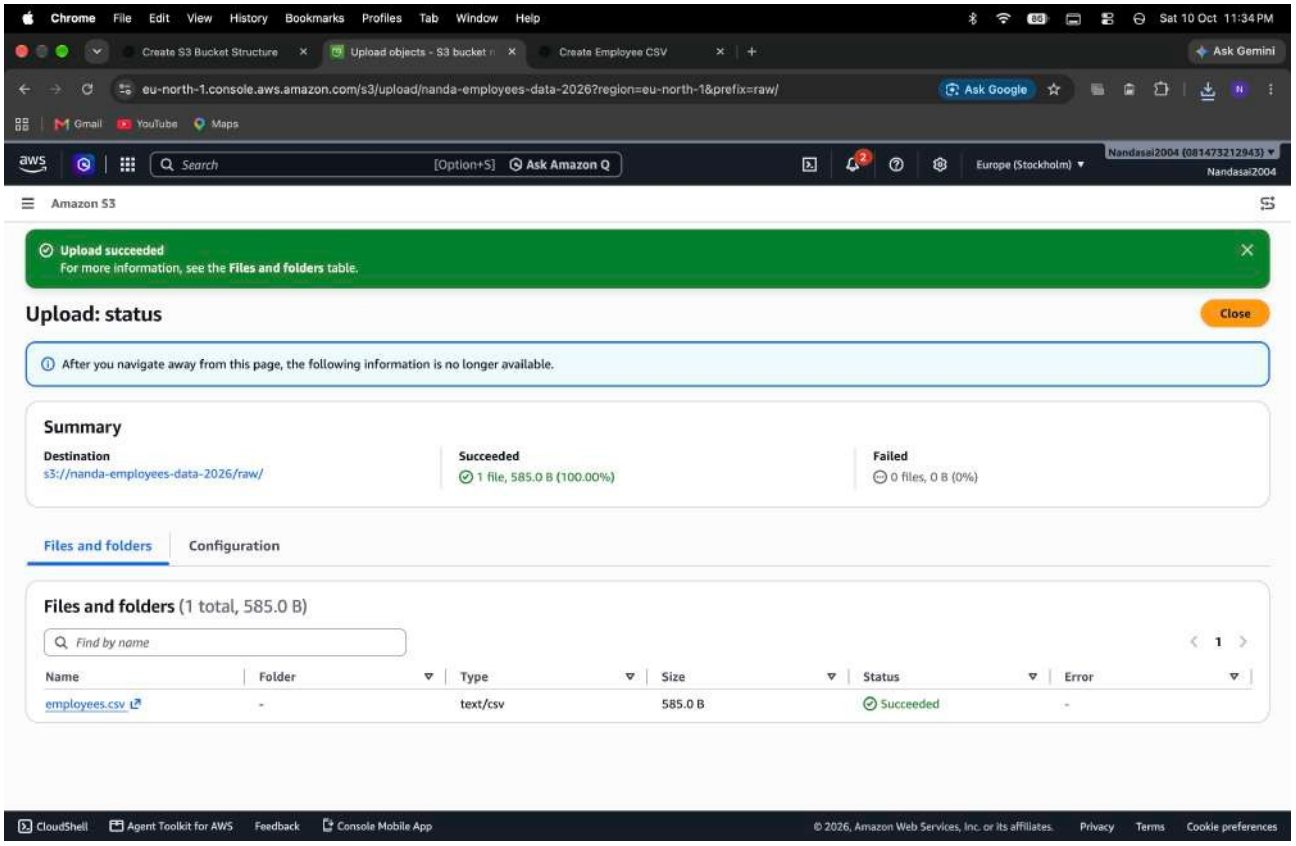
Selected AWS Region



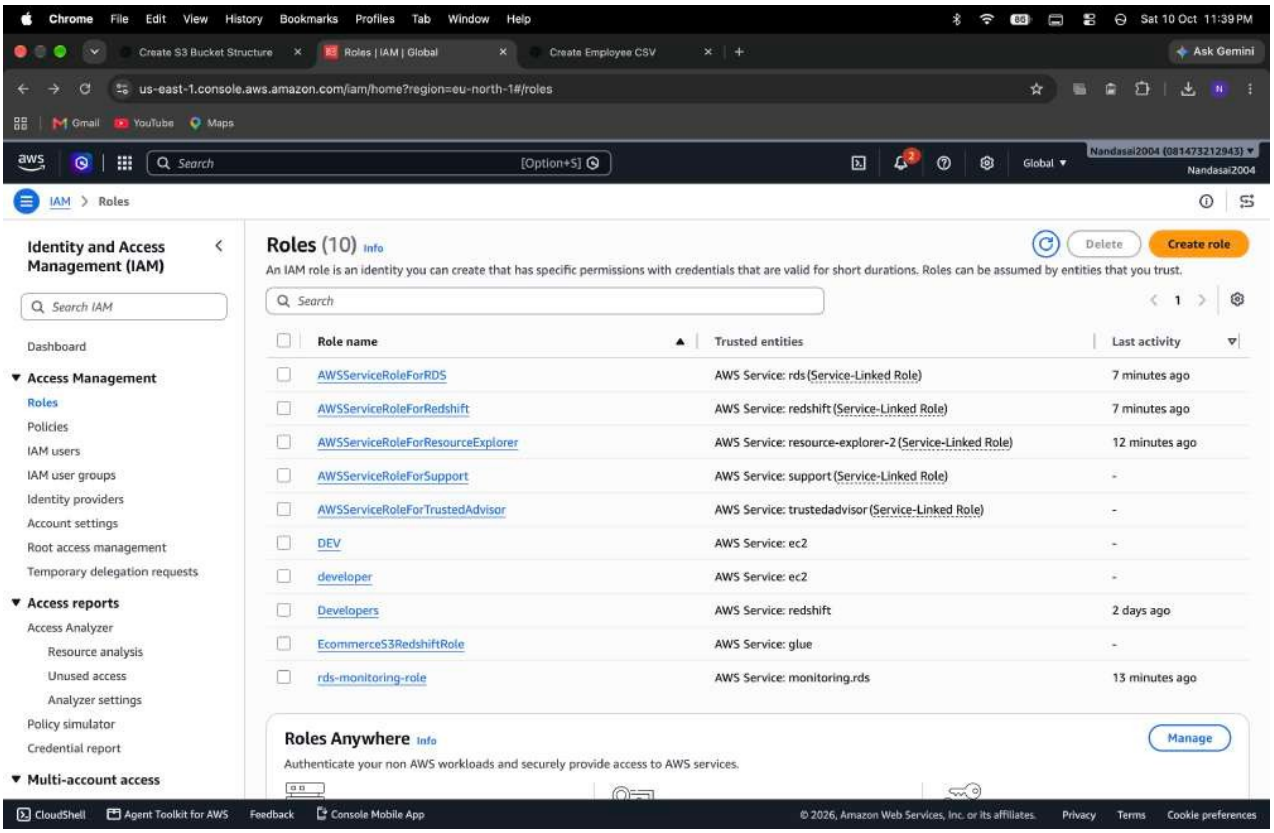
Amazon S3:

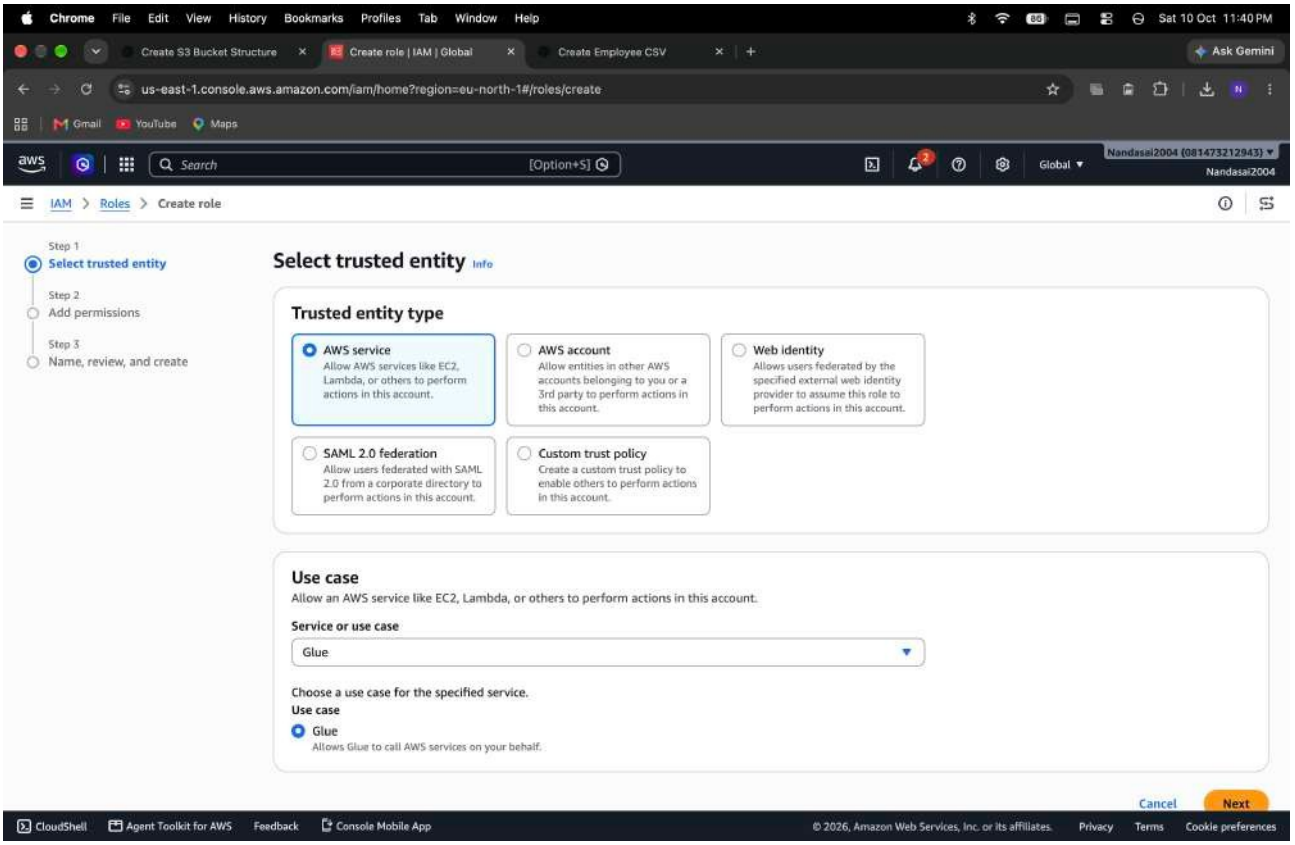

Amazon S3
Buckets
General purpose buckets
Directory buckets
General purpose buckets (3) Info
Copy ARN
Empty
Delete
Create bucket
Buckets are containers for data stored in S3.
Find buckets by name
Name
AWS Region
Creation date
ecommerce-pratice
Europe (Stockholm) eu-north-1
October 9, 2026, 09:57:03 (UTC+05:30)
my-pratice-1
Europe (Stockholm) eu-north-1
October 8, 2026, 09:52:24 (UTC+05:30)
nandasai-1908
Europe (Stockholm) eu-north-1
October 7, 2026, 10:26:53 (UTC+05:30)
Account snapshot Info
Updated daily
View dashboard
Storage Lens provides visibility into storage usage and activity trends.
External access summary Info
Updated daily
External access findings help you identify bucket permissions that allow public access or access from other AWS accounts.

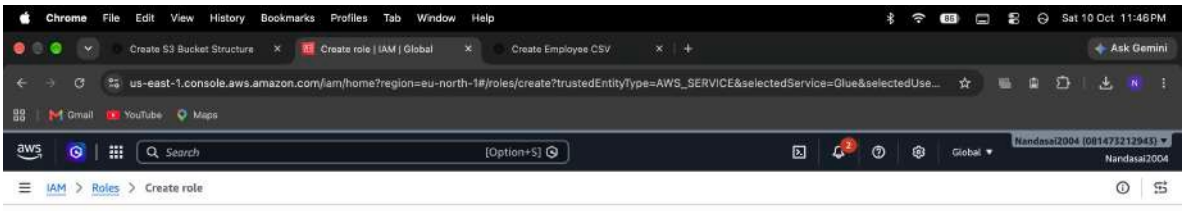

Amazon S3 > Buckets > nanda-employees-data-2026
Successfully created folder "raw".
nanda-employees-data-2026 Info
Objects
Metadata
Properties
Permissions
Metrics
Management
File systems
Access Points
Objects (1)
Copy S3 URI
Copy URL
Download
Open
Delete
Actions
Create folder
Upload
Objects are the fundamental entities stored in Amazon S3. You can use Amazon S3 inventory to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. Learn more
Find objects by prefix
Name
Type
Last modified
Size
Storage class
raw/
Folder

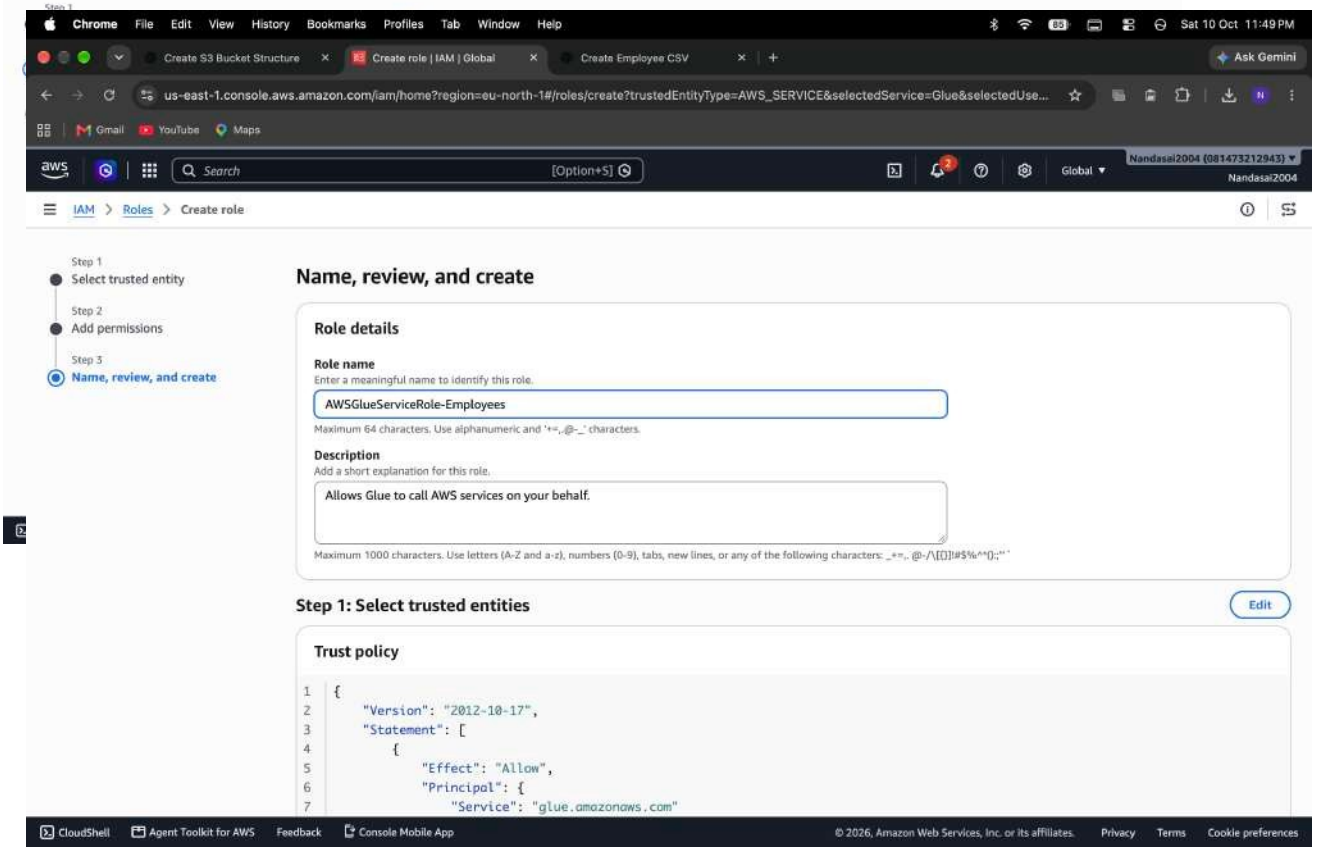

Upload succeeded
For more information, see the Files and folders table.
Upload: status
Close
After you navigate away from this page, the following information is no longer available.
Summary
Destination
s3://nanda-employees-data-2026/raw/
Succeeded
1 file, 585.0 B (100.00%)
Failed
0 files, 0 B (0%)
Files and folders
Configuration
Files and folders (1 total, 585.0 B)
Find by name
Name
Folder
Type
Size
Status
Error
employees.csv
text/csv
585.0 B
Succeeded

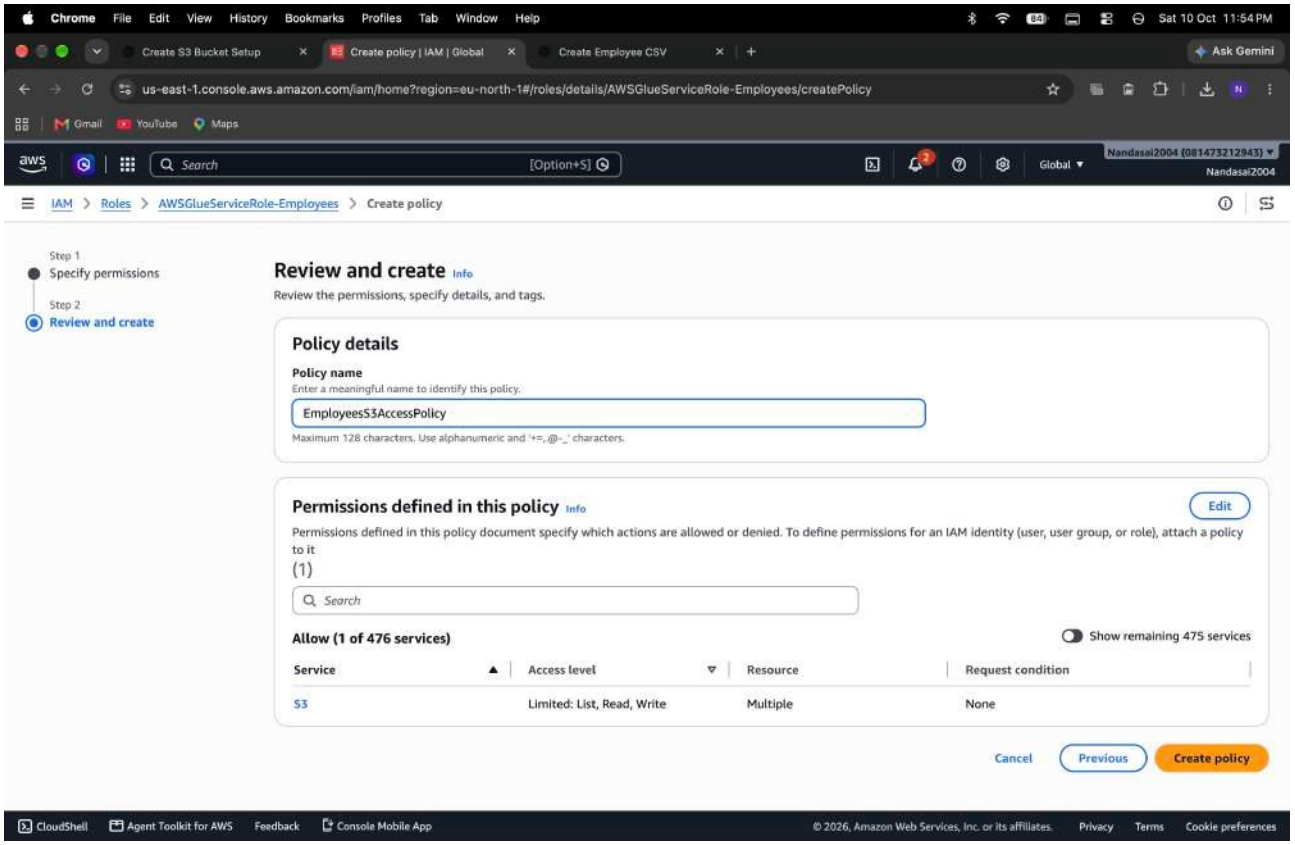
IAM

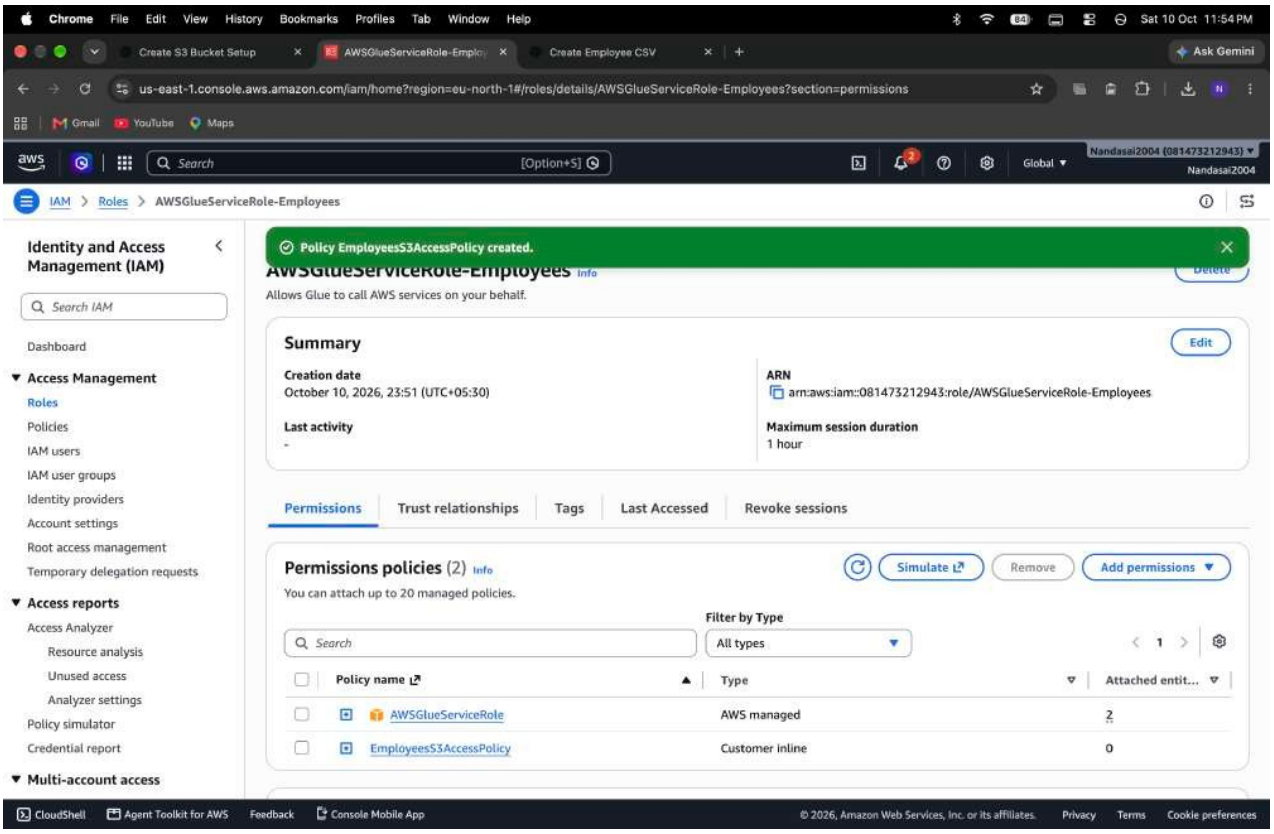

Roles (10) Info
An IAM role is an identity you can create that has specific permissions with credentials that are valid for short durations. Roles can be assumed by entities that you trust.
Delete
Create role
Search
Role name
Trusted entities
Last activity
AWSServiceRoleForRDS
AWS Service: rds (Service-Linked Role)
7 minutes ago
AWSServiceRoleForRedshift
AWS Service: redshift (Service-Linked Role)
7 minutes ago
AWSServiceRoleForResourceExplorer
AWS Service: resource-explorer-2 (Service-Linked Role)
12 minutes ago
AWSServiceRoleForSupport
AWS Service: support (Service-Linked Role)
AWSServiceRoleForTrustedAdvisor
AWS Service: trustedadvisor (Service-Linked Role)
DEV
AWS Service: ec2
developer
AWS Service: ec2
Developers
AWS Service: redshift
2 days ago
EcommerceS3RedshiftRole
AWS Service: glue
rds-monitoring-role
AWS Service: monitoring.rds
13 minutes ago
Roles Anywhere Info
Manage
Authenticate your non AWS workloads and securely provide access to AWS services.


Chrome File Edit View History Bookmarks Profiles Tab Window Help
Sat 10 Oct 11:40 PM
Create S3 Bucket Structure
Create role | IAM | Global
Create Employee CSV
Ask Gemini
us-east-1.console.aws.amazon.com/iam/home?region=eu-north-1#/roles/create
Gmail YouTube Maps
Search [Option+S]
Global
Nandasai2004 (081473212943)
Nandasai2004
IAM > Roles > Create role
Step 1
Select trusted entity
Step 2
Add permissions
Step 3
Name, review, and create
Select trusted entity Info
Trusted entity type
AWS service
Allow AWS services like EC2, Lambda, or others to perform actions in this account.
AWS account
Allow entities in other AWS accounts belonging to you or a 3rd party to perform actions in this account.
Web identity
Allows users federated by the specified external web identity provider to assume this role to perform actions in this account.
SAML 2.0 federation
Allow users federated with SAML 2.0 from a corporate directory to perform actions in this account.
Custom trust policy
Create a custom trust policy to enable others to perform actions in this account.
Use case
Allow an AWS service like EC2, Lambda, or others to perform actions in this account.
Service or use case
Glue
Choose a use case for the specified service.
Use case
Glue
Allows Glue to call AWS services on your behalf.
Cancel
Next
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
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Terms
Cookie preferences


Chrome File Edit View History Bookmarks Profiles Tab Window Help
Sat 10 Oct 11:46 PM
Create S3 Bucket Structure
Create role | IAM | Global
Create Employee CSV
Ask Gemini
us-east-1.console.aws.amazon.com/iam/home?region=eu-north-1#/roles/create?trustedEntityType=AWS_SERVICE&selectedService=Glue&selectedUse...
Gmail YouTube Maps
Search [Option+S]
Global
Nandasai2004 (081473212943)
Nandasai2004
IAM > Roles > Create role

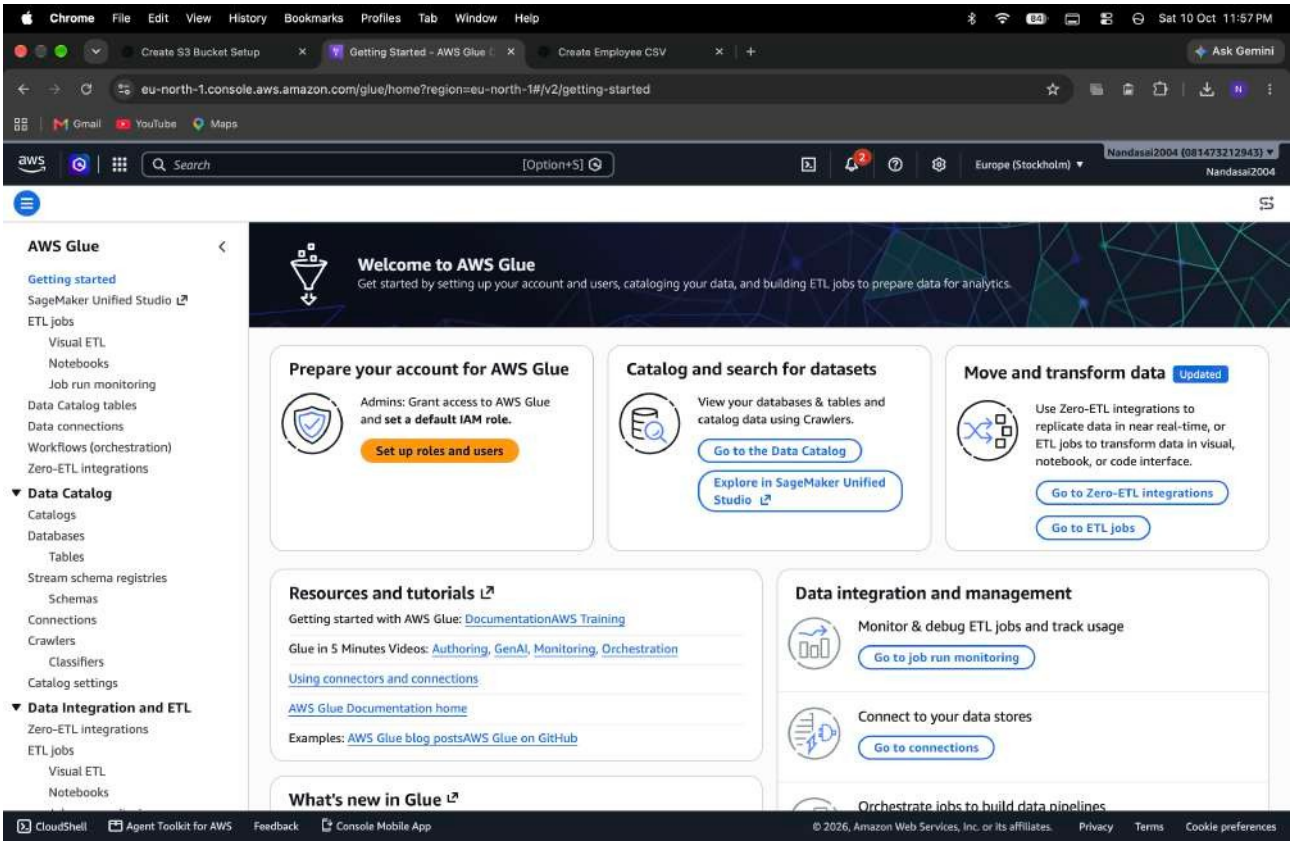

Chrome File Edit View History Bookmarks Profiles Tab Window Help
Sat 10 Oct 11:49 PM
Create S3 Bucket Structure
Create role | IAM | Global
Create Employee CSV
Ask Gemini
us-east-1.console.aws.amazon.com/iam/home?region=eu-north-1#/roles/create?trustedEntityType=AWS_SERVICE&selectedService=Glue&selectedUse...
Gmail YouTube Maps
Search [Option+S]
Global
Nandasai2004 (081473212943)
Nandasai2004
IAM > Roles > Create role
Step 1
Select trusted entity
Step 2
Add permissions
Step 3
Name, review, and create
Name, review, and create
Role details
Role name
Enter a meaningful name to identify this role.
AWSGlueServiceRole-Employees
Maximum 64 characters. Use alphanumeric and '+=,.@-_' characters.
Description
Add a short explanation for this role.
Allows Glue to call AWS services on your behalf.
Maximum 1000 characters. Use letters (A-Z and a-z), numbers (0-9), tabs, new lines, or any of the following characters: _+=,. @-/\[{}]!#$%^*();:"'`
Step 1: Select trusted entities
Edit
Trust policy
{
"Version": "2012-10-17",
"Statement": [
{
"Effect": "Allow",
"Principal": {
"Service": "glue.amazonaws.com"
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
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Terms
Cookie preferences


Chrome File Edit View History Bookmarks Profiles Tab Window Help
Sat 10 Oct 11:54 PM
Create S3 Bucket Setup
Create policy | IAM | Global
Create Employee CSV
Ask Gemini
us-east-1.console.aws.amazon.com/iam/home?region=eu-north-1#/roles/details/AWSGlueServiceRole-Employees/createPolicy
Gmail YouTube Maps
aws
Search [Option+S]
Global
Nandasai2004 (081473212943)
Nandasai2004
IAM > Roles > AWSGlueServiceRole-Employees > Create policy
Step 1
Specify permissions
Step 2
Review and create
Review and create Info
Review the permissions, specify details, and tags.
Policy details
Policy name
Enter a meaningful name to identify this policy.
EmployeesS3AccessPolicy
Maximum 128 characters. Use alphanumeric and '+=,.@-_' characters.
Permissions defined in this policy Info
Edit
Permissions defined in this policy document specify which actions are allowed or denied. To define permissions for an IAM identity (user, user group, or role), attach a policy to it
(1)
Search
Allow (1 of 476 services)
Show remaining 475 services
Service
Access level
Resource
Request condition
S3
Limited: List, Read, Write
Multiple
None
Cancel
Previous
Create policy
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
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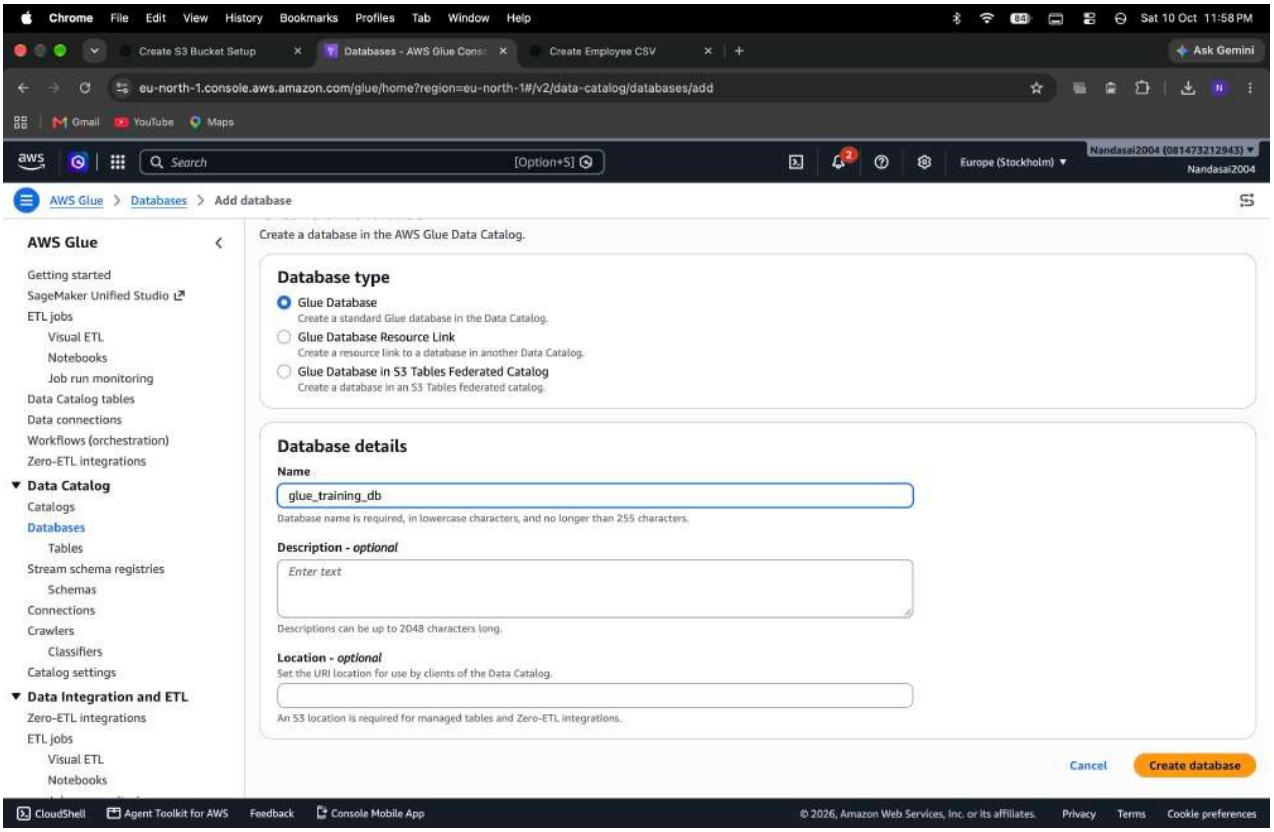

Chrome File Edit View History Bookmarks Profiles Tab Window Help
Sat 10 Oct 11:54 PM
Create S3 Bucket Setup
AWSGlueServiceRole-Employ
Create Employee CSV
Ask Gemini
us-east-1.console.aws.amazon.com/iam/home?region=eu-north-1#/roles/details/AWSGlueServiceRole-Employees?section=permissions
Gmail YouTube Maps
aws
Search [Option+S]
Global
Nandasai2004 (081473212943)
Nandasai2004
IAM > Roles > AWSGlueServiceRole-Employees
Identity and Access Management (IAM)
Search IAM
Dashboard
Access Management
Roles
Policies
IAM users
IAM user groups
Identity providers
Account settings
Root access management
Temporary delegation requests
Access reports
Access Analyzer
Resource analysis
Unused access
Analyzer settings
Policy simulator
Credential report
Multi-account access
Policy EmployeesS3AccessPolicy created.
AWSGlueServiceRole-Employees Info
Delete
Allows Glue to call AWS services on your behalf.
Summary
Edit
Creation date
October 10, 2026, 23:51 (UTC+05:30)
ARN
arn:aws:iam::081473212943:role/AWSGlueServiceRole-Employees
Last activity
-
Maximum session duration
1 hour
Permissions
Trust relationships
Tags
Last Accessed
Revoke sessions
Permissions policies (2) Info
You can attach up to 20 managed policies.
Simulate
Remove
Add permissions
Filter by Type
Search
All types
1
Policy name
Type
Attached entit...
AWSGlueServiceRole
AWS managed
2
EmployeesS3AccessPolicy
Customer inline
0
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
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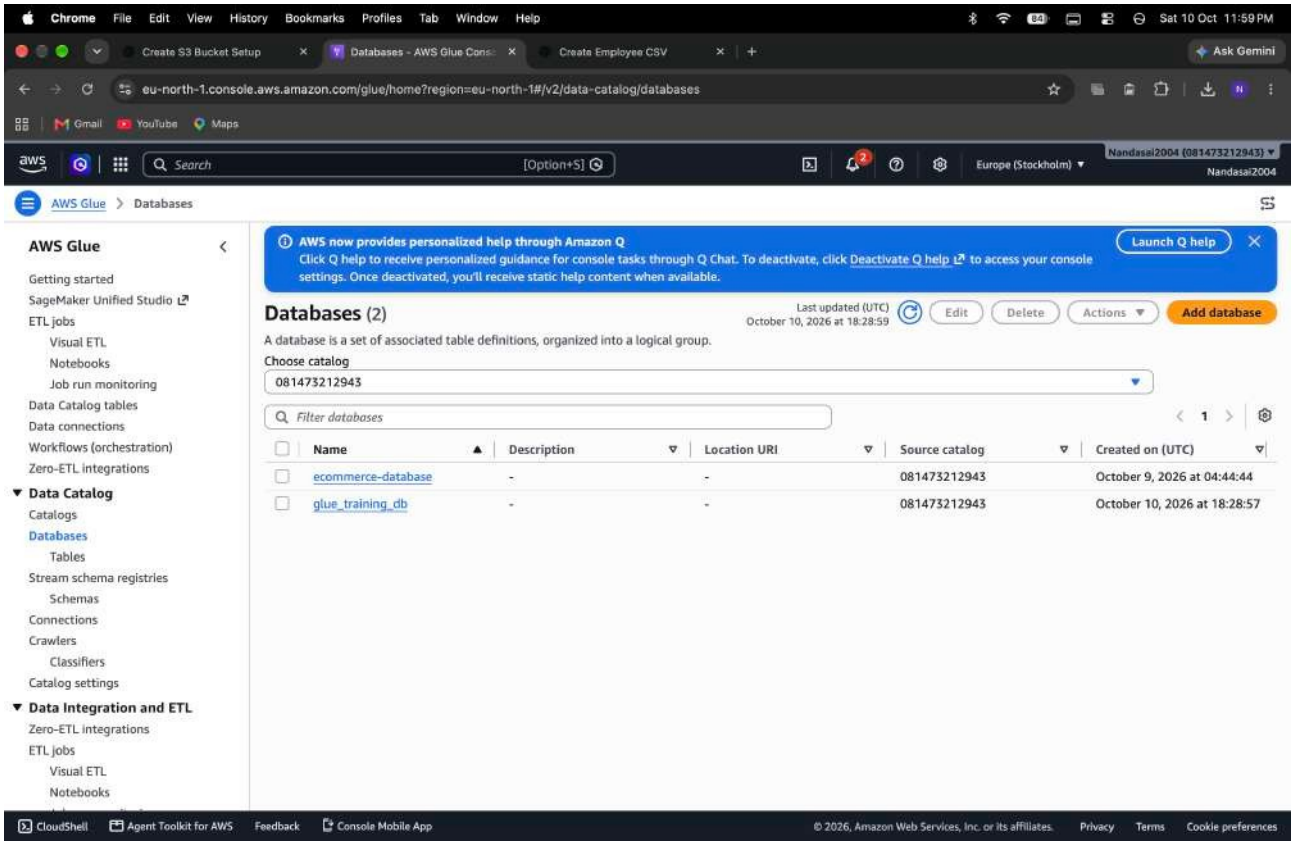
AWS Glue Database

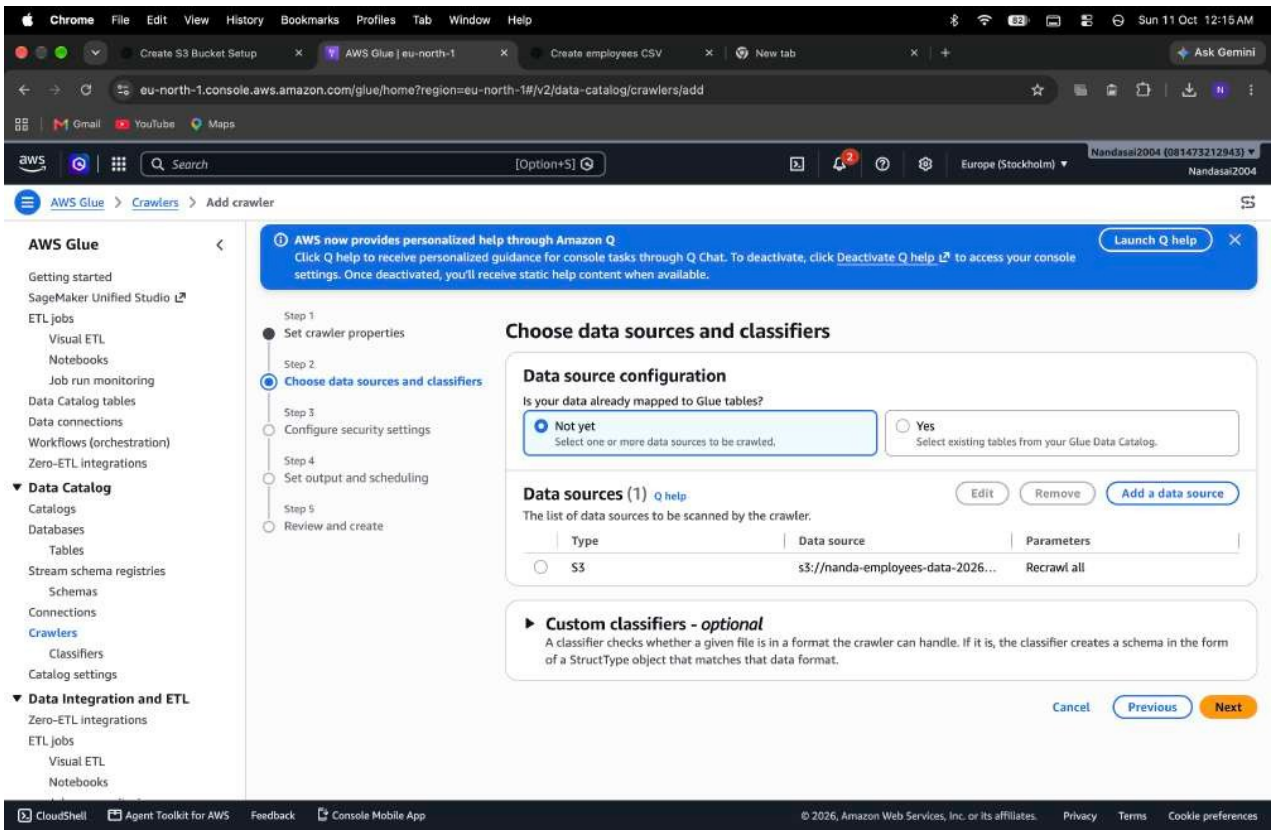
AWS Glue Console

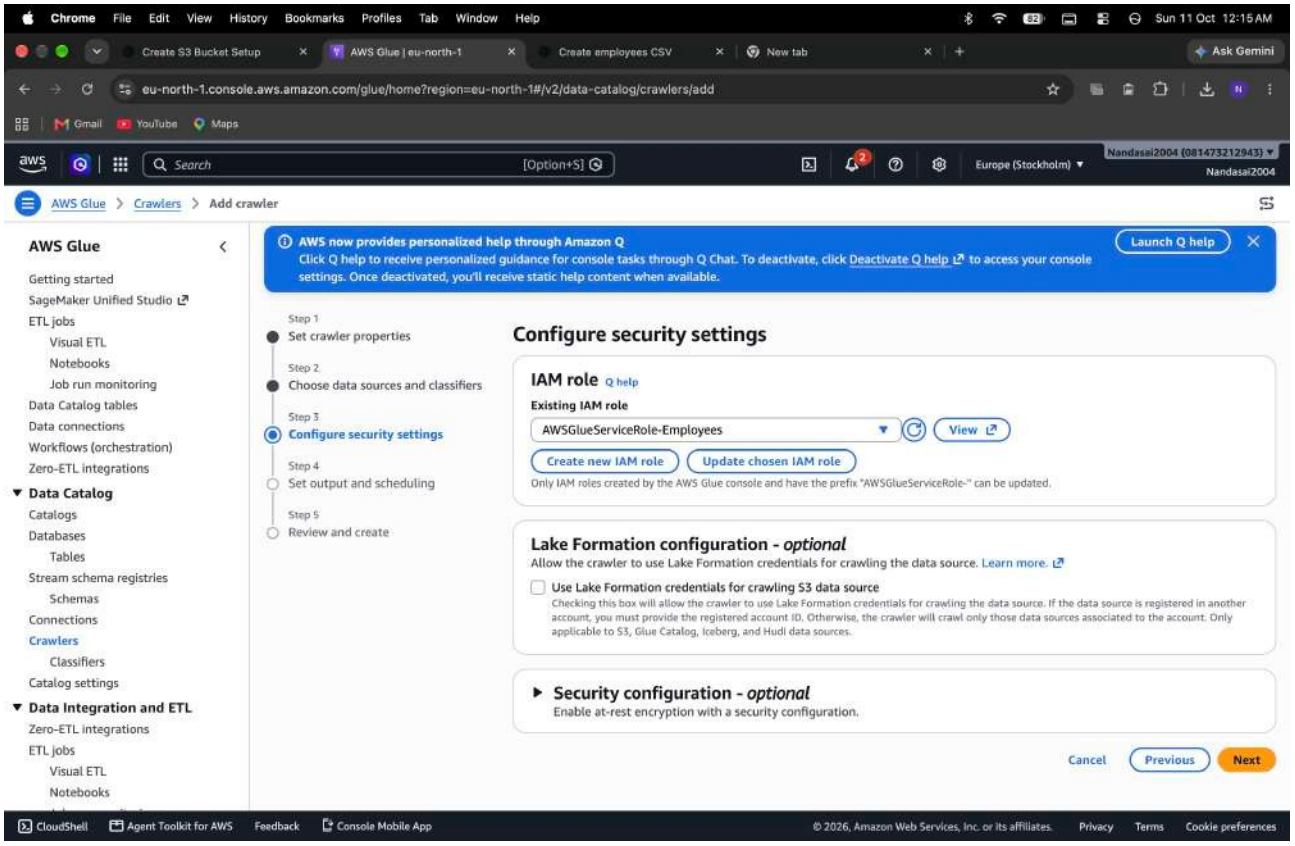


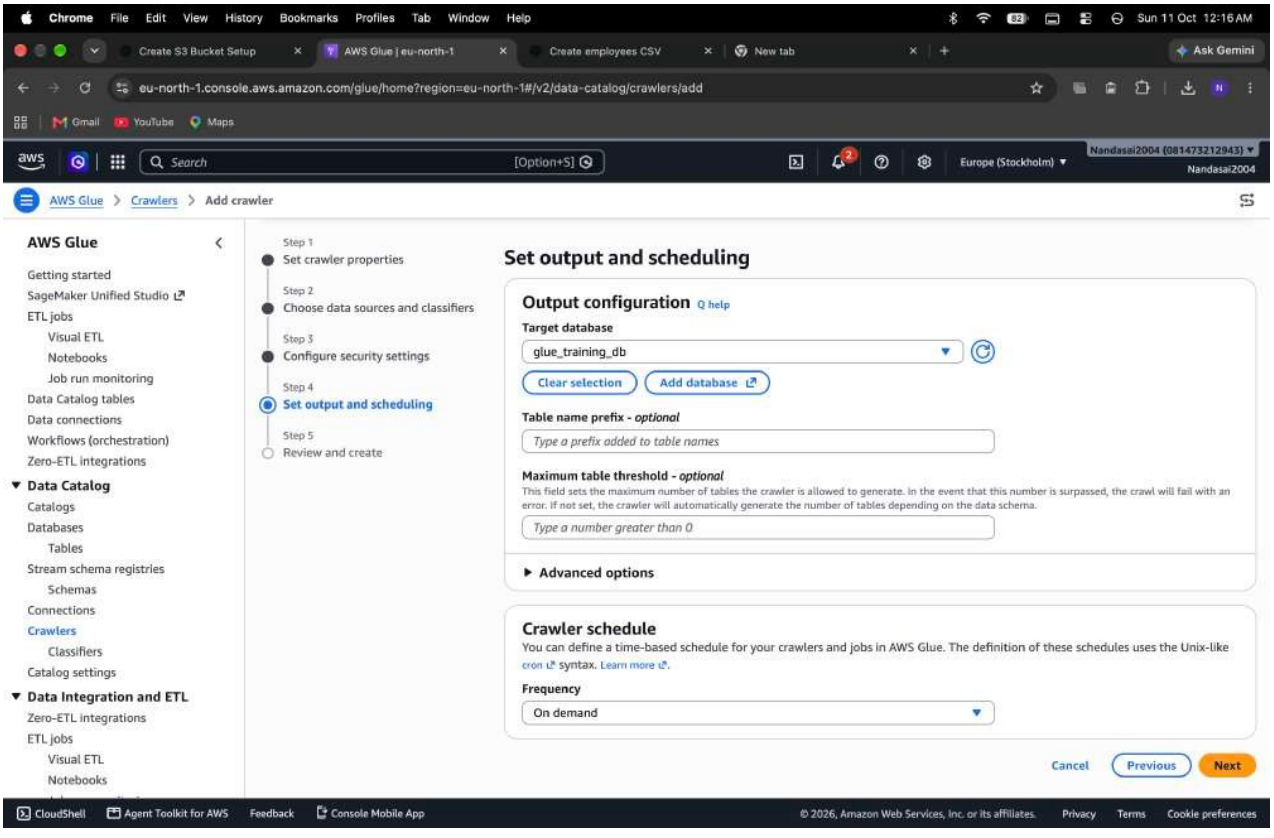
Data Catalog:

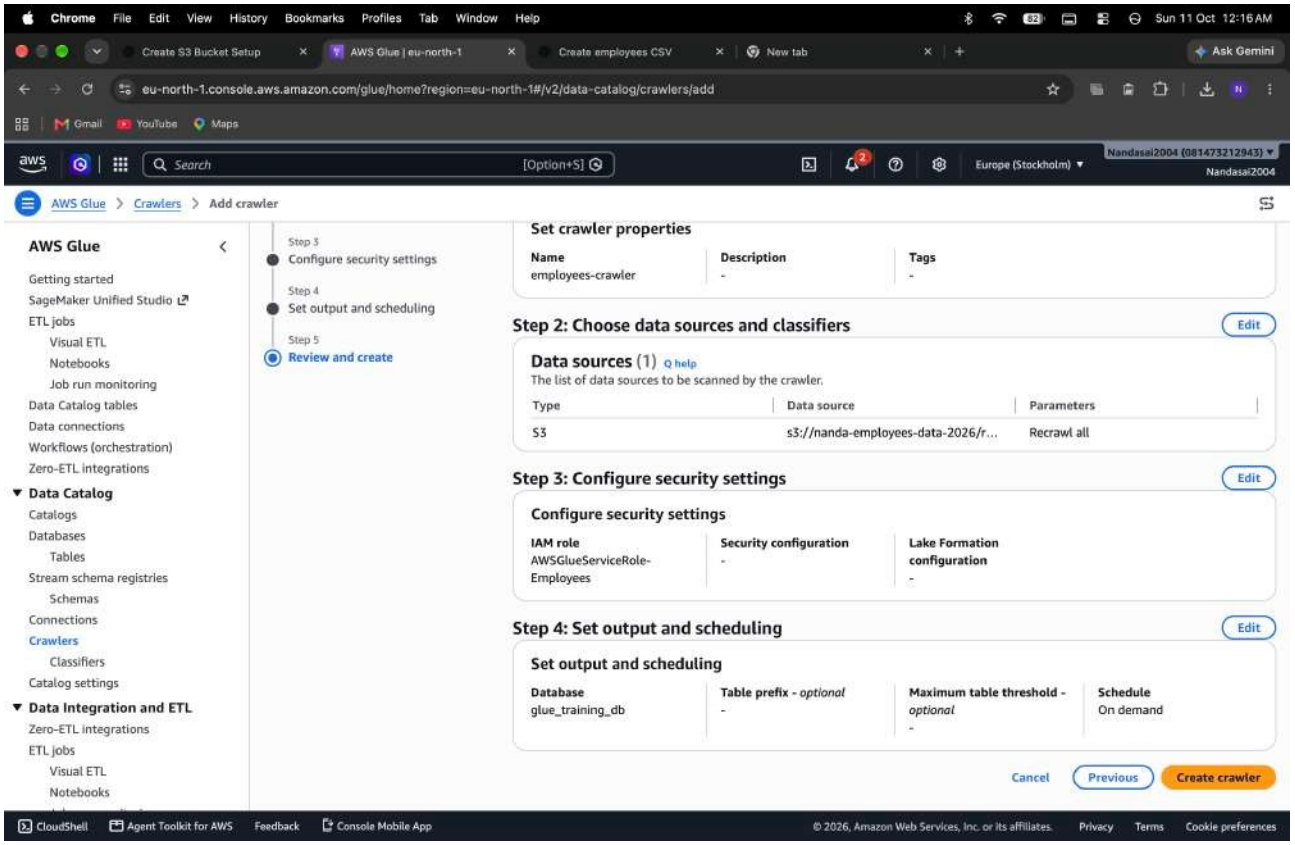


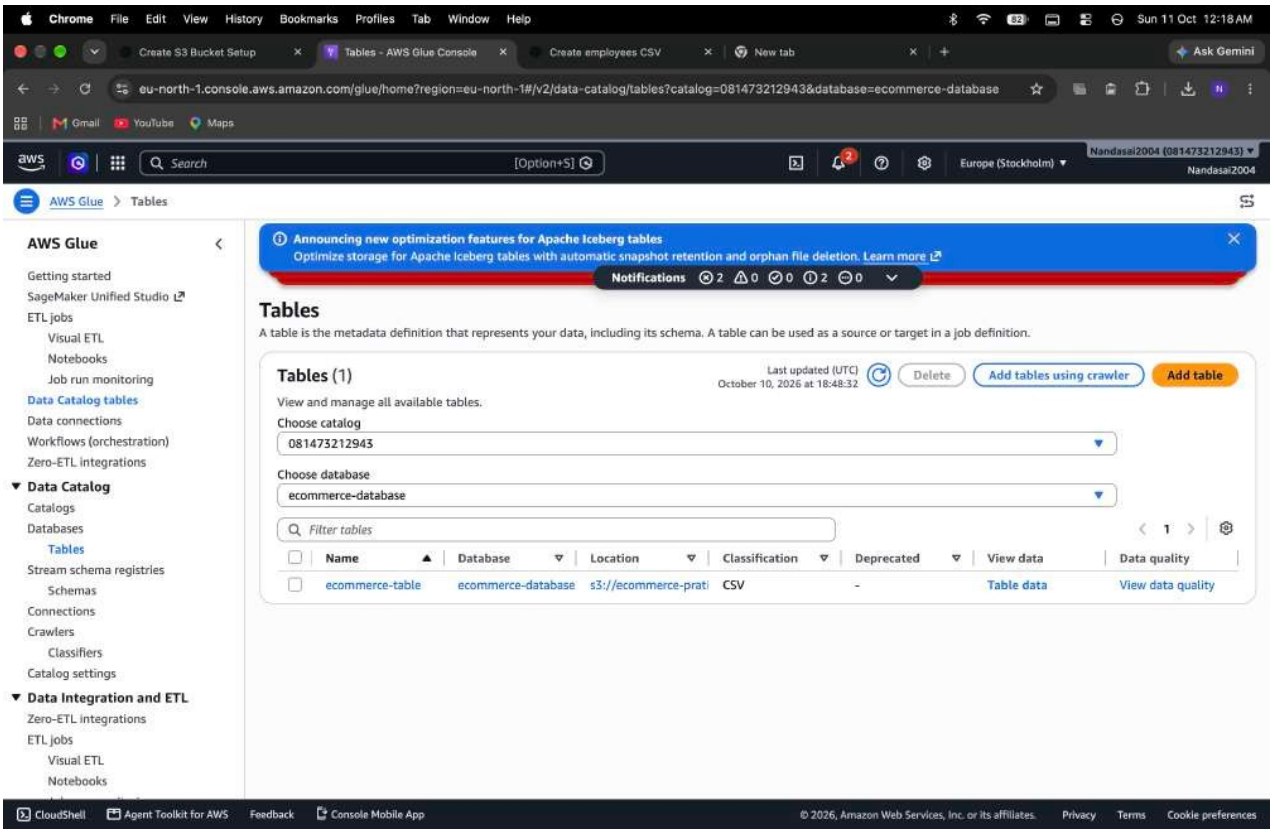

Chrome File Edit View History Bookmarks Profiles Tab Window Help
Sat 10 Oct 11:59 PM
Create S3 Bucket Setup
Databases - AWS Glue Cons
Create Employee CSV
Ask Gemini
eu-north-1.console.aws.amazon.com/glue/home?region=eu-north-1#/v2/data-catalog/databases
Gmail YouTube Maps
Search [Option+S]
Europe (Stockholm)
Nandasai2004 (081473212943)
Nandasai2004
AWS Glue > Databases
AWS Glue
Getting started
SageMaker Unified Studio
ETL jobs
Visual ETL
Notebooks
Job run monitoring
Data Catalog tables
Data connections
Workflows (orchestration)
Zero-ETL integrations
Data Catalog
Catalogs
Databases
Tables
Stream schema registries
Schemas
Connections
Crawlers
Classifiers
Catalog settings
Data Integration and ETL
Zero-ETL integrations
ETL jobs
Visual ETL
Notebooks
AWS now provides personalized help through Amazon Q
Click Q help to receive personalized guidance for console tasks through Q Chat. To deactivate, click Deactivate Q help to access your console settings. Once deactivated, you'll receive static help content when available.
Launch Q help
Databases (2)
Last updated (UTC) October 10, 2026 at 18:28:59
Edit Delete Actions Add database
A database is a set of associated table definitions, organized into a logical group.
Choose catalog
081473212943
Filter databases
1
Name Description Location URI Source catalog Created on (UTC)
ecommerce-database - - 081473212943 October 9, 2026 at 04:44:44
glue_training_db - - 081473212943 October 10, 2026 at 18:28:57
CloudShell Agent Toolkit for AWS Feedback Console Mobile App
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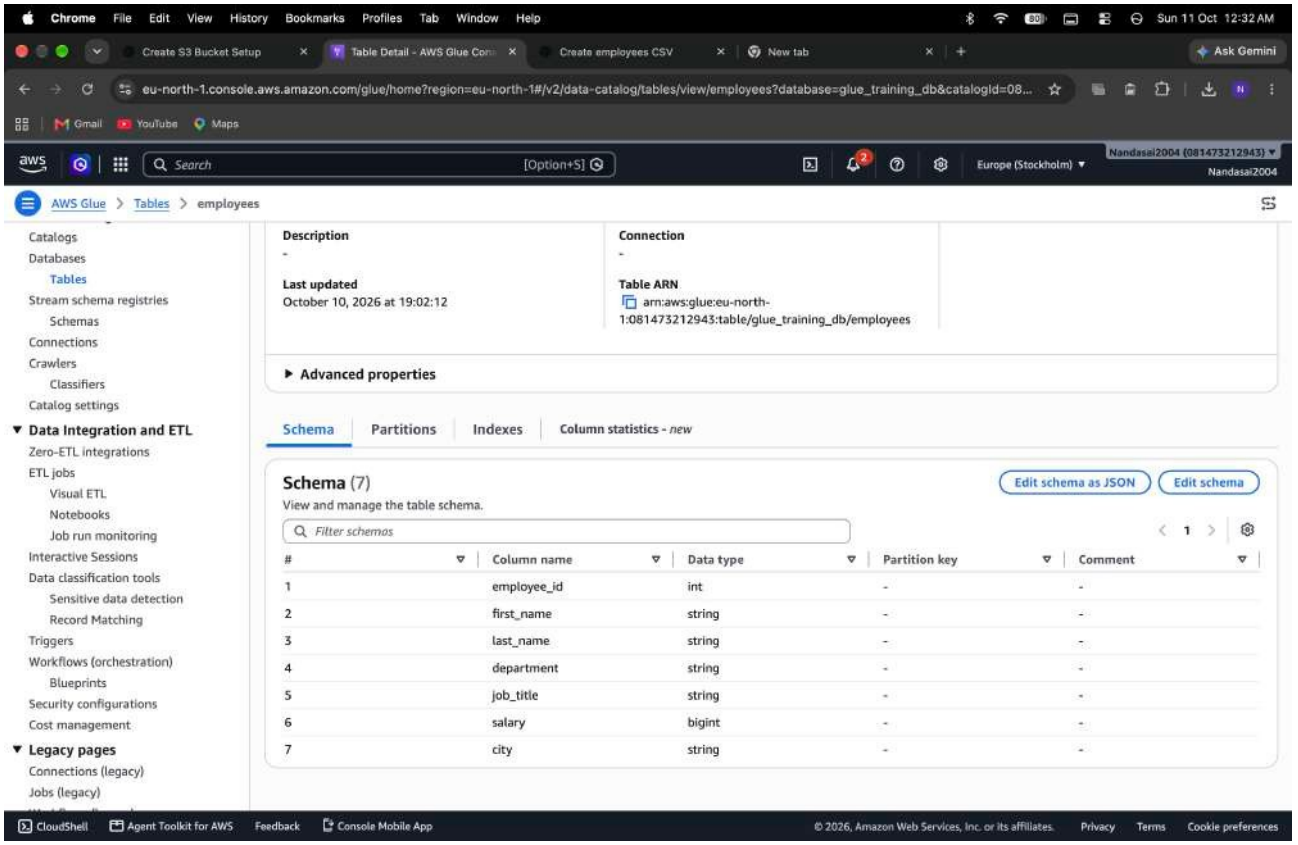

Chrome File Edit View History Bookmarks Profiles Tab Window Help
Sun 11 Oct 12:15 AM
Create S3 Bucket Setup
AWS Glue | eu-north-1
Create employees CSV
New tab
Ask Gemini
eu-north-1.console.aws.amazon.com/glue/home?region=eu-north-1#/v2/data-catalog/crawlers/add
Gmail YouTube Maps
Search [Option+S]
Europe (Stockholm)
Nandasai2004 (081473212943)
Nandasai2004
AWS Glue > Crawlers > Add crawler
AWS Glue
Getting started
SageMaker Unified Studio
ETL jobs
Visual ETL
Notebooks
Job run monitoring
Data Catalog tables
Data connections
Workflows (orchestration)
Zero-ETL integrations
Data Catalog
Catalogs
Databases
Tables
Stream schema registries
Schemas
Connections
Crawlers
Classifiers
Catalog settings
Data Integration and ETL
Zero-ETL integrations
ETL jobs
Visual ETL
Notebooks
AWS now provides personalized help through Amazon Q
Click Q help to receive personalized guidance for console tasks through Q Chat. To deactivate, click Deactivate Q help to access your console settings. Once deactivated, you'll receive static help content when available.
Launch Q help
Step 1 Set crawler properties
Step 2 Choose data sources and classifiers
Step 3 Configure security settings
Step 4 Set output and scheduling
Step 5 Review and create
Choose data sources and classifiers
Data source configuration
Is your data already mapped to Glue tables?
Not yet
Select one or more data sources to be crawled.
Yes
Select existing tables from your Glue Data Catalog.
Data sources (1) Q help
Edit Remove Add a data source
The list of data sources to be scanned by the crawler.
Type Data source Parameters
S3 s3://nanda-employees-data-2026... Recrawl all
Custom classifiers - optional
A classifier checks whether a given file is in a format the crawler can handle. If it is, the classifier creates a schema in the form of a StructType object that matches that data format.
Cancel Previous Next
CloudShell Agent Toolkit for AWS Feedback Console Mobile App
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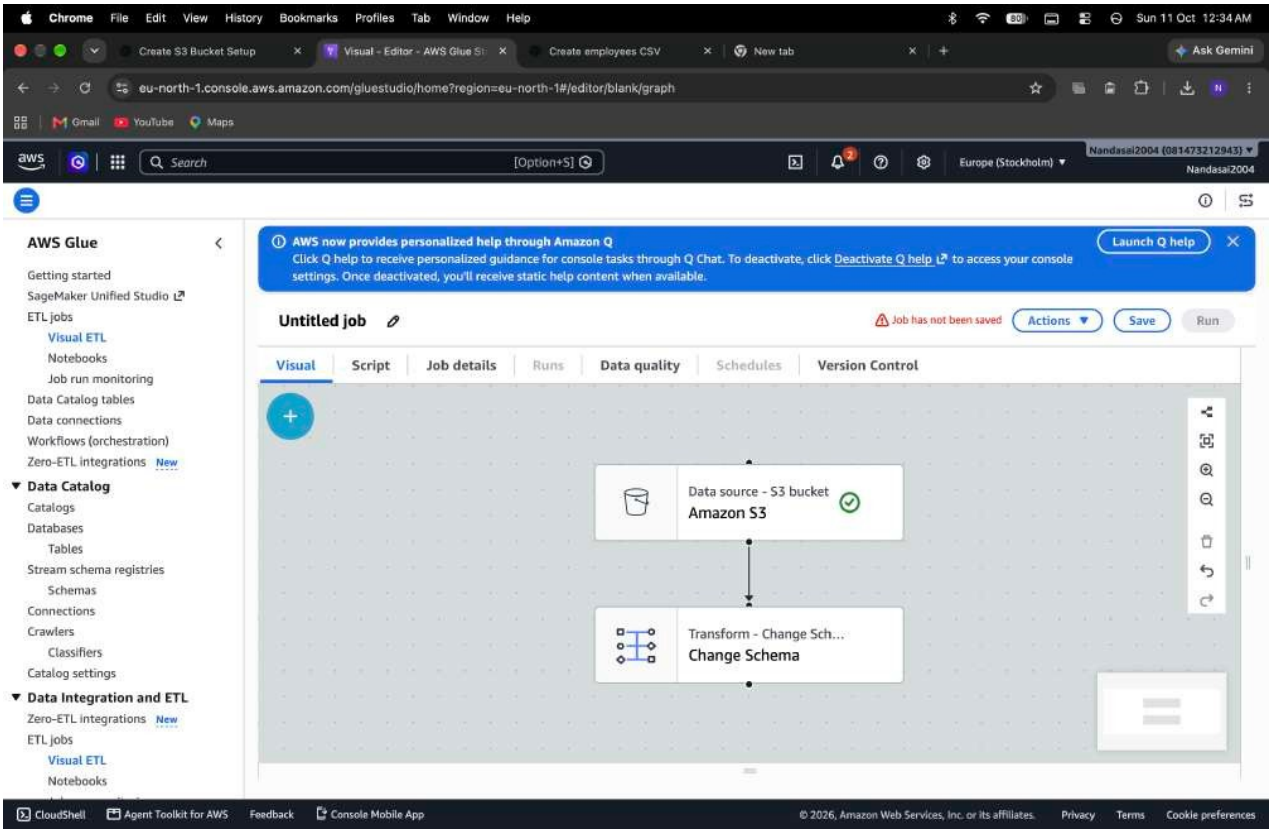

Chrome File Edit View History Bookmarks Profiles Tab Window Help
Sun 11 Oct 12:15 AM
Create S3 Bucket Setup
AWS Glue | eu-north-1
Create employees CSV
New tab
Ask Gemini
eu-north-1.console.aws.amazon.com/glue/home?region=eu-north-1#/v2/data-catalog/crawlers/add
Gmail YouTube Maps
Search [Option+S]
Europe (Stockholm)
Nandasai2004 (081473212943)
Nandasai2004
AWS Glue > Crawlers > Add crawler
AWS Glue
Getting started
SageMaker Unified Studio
ETL jobs
Visual ETL
Notebooks
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AWS now provides personalized help through Amazon Q
Click Q help to receive personalized guidance for console tasks through Q Chat. To deactivate, click Deactivate Q help to access your console settings. Once deactivated, you'll receive static help content when available.
Launch Q help
Step 1
Set crawler properties
Step 2
Choose data sources and classifiers
Step 3
Configure security settings
Step 4
Set output and scheduling
Step 5
Review and create
Configure security settings
IAM role Q help
Existing IAM role
AWSGlueServiceRole-Employees
View
Create new IAM role
Update chosen IAM role
Only IAM roles created by the AWS Glue console and have the prefix "AWSGlueServiceRole-" can be updated.
Lake Formation configuration - optional
Allow the crawler to use Lake Formation credentials for crawling the data source. Learn more.
Use Lake Formation credentials for crawling S3 data source
Checking this box will allow the crawler to use Lake Formation credentials for crawling the data source. If the data source is registered in another account, you must provide the registered account ID. Otherwise, the crawler will crawl only those data sources associated to the account. Only applicable to S3, Glue Catalog, Iceberg, and Hudi data sources.
Security configuration - optional
Enable at-rest encryption with a security configuration.
Cancel
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Chrome File Edit View History Bookmarks Profiles Tab Window Help
Sun 11 Oct 12:16 AM
Create S3 Bucket Setup
AWS Glue | eu-north-1
Create employees CSV
New tab
Ask Gemini
eu-north-1.console.aws.amazon.com/glue/home?region=eu-north-1#/v2/data-catalog/crawlers/add
Gmail YouTube Maps
Search [Option+S]
Europe (Stockholm)
Nandasai2004 (081473212943)
Nandasai2004
AWS Glue > Crawlers > Add crawler
AWS Glue
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Step 1
Set crawler properties
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Step 5
Review and create
Set output and scheduling
Output configuration Q help
Target database
glue_training_db
Clear selection
Add database
Table name prefix - optional
Type a prefix added to table names
Maximum table threshold - optional
This field sets the maximum number of tables the crawler is allowed to generate. In the event that this number is surpassed, the crawl will fail with an error. If not set, the crawler will automatically generate the number of tables depending on the data schema.
Type a number greater than 0
Advanced options
Crawler schedule
You can define a time-based schedule for your crawlers and jobs in AWS Glue. The definition of these schedules uses the Unix-like cron syntax. Learn more.
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Create S3 Bucket Setup
AWS Glue | eu-north-1
Create employees CSV
New tab
eu-north-1.console.aws.amazon.com/glue/home?region=eu-north-1#/v2/data-catalog/crawlers/add
Search
Europe (Stockholm)
Nandasai2004
AWS Glue > Crawlers > Add crawler
Step 3
Configure security settings
Step 4
Set output and scheduling
Step 5
Review and create
Set crawler properties
Name
employees-crawler
Description
Tags
Step 2: Choose data sources and classifiers
Edit
Data sources (1)
The list of data sources to be scanned by the crawler.
Type
Data source
Parameters
S3
s3://nanda-employees-data-2026/r...
Recrawl all
Step 3: Configure security settings
Configure security settings
IAM role
AWSGlueServiceRole-Employees
Security configuration
Lake Formation configuration
Step 4: Set output and scheduling
Set output and scheduling
Database
glue_training_db
Table prefix - optional
Maximum table threshold - optional
Schedule
On demand
Cancel
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Tables - AWS Glue Console
eu-north-1.console.aws.amazon.com/glue/home?region=eu-north-1#/v2/data-catalog/tables?catalog=081473212943&database=ecommerce-database
AWS Glue > Tables
Announcing new optimization features for Apache Iceberg tables
Optimize storage for Apache Iceberg tables with automatic snapshot retention and orphan file deletion. Learn more
Tables
A table is the metadata definition that represents your data, including its schema. A table can be used as a source or target in a job definition.
Tables (1)
Last updated (UTC)
October 10, 2026 at 18:48:32
Delete
Add tables using crawler
Add table
View and manage all available tables.
Choose catalog
081473212943
Choose database
ecommerce-database
Filter tables
Name
Database
Location
Classification
Deprecated
View data
Data quality
ecommerce-table
ecommerce-database
s3://ecommerce-prati
CSV
-
Table data
View data quality


Chrome File Edit View History Bookmarks Profiles Tab Window Help
Sun 11 Oct 12:32 AM
Create S3 Bucket Setup
Table Detail - AWS Glue Con
Create employees CSV
New tab
Ask Gemini
eu-north-1.console.aws.amazon.com/glue/home?region=eu-north-1#/v2/data-catalog/tables/view/employees?database=glue_training_db&catalogId=08...
Gmail YouTube Maps
Search [Option+S]
Europe (Stockholm)
Nandasai2004 (081473212943)
Nandasai2004
AWS Glue > Tables > employees
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Blueprints
Security configurations
Cost management
Legacy pages
Connections (legacy)
Jobs (legacy)
Description
-
Last updated
October 10, 2026 at 19:02:12
Connection
-
Table ARN
arn:aws:glue:eu-north-1:081473212943:table/glue_training_db/employees
Advanced properties
Schema
Partitions
Indexes
Column statistics - new
Schema (7)
View and manage the table schema.
Edit schema as JSON
Edit schema
Filter schemas
1
Column name Data type Partition key Comment
1 employee_id int - -
2 first_name string - -
3 last_name string - -
4 department string - -
5 job_title string - -
6 salary bigint - -
7 city string - -
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Visual - Editor - AWS Glue St
Create employees CSV
New tab
Ask Gemini
eu-north-1.console.aws.amazon.com/gluestudio/home?region=eu-north-1#/editor/blank/graph
Gmail YouTube Maps
Search [Option+S]
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Launch Q help
Untitled job
Job has not been saved
Actions
Save
Run
Visual
Script
Job details
Runs
Data quality
Schedules
Version Control
Data source - S3 bucket
Amazon S3
Transform - Change Sch...
Change Schema
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1) Amazon S3 (Raw data) :-

=> i) upload employees.csv file to S3 in a folder called raw

=> this file contains employees details like employee_id, name, department, salary and city.

2) AWS Glue Crawler :-

=> It automatically detects the columns and their data types

=> It creates a table in Glue data catalog

3) AWS Glue data catalog :-

=> The metadata of the employees data is stored in a database called glue-training-db;

=> It creates a table named employees with column information, data types and with the S3 location

4) AWS Glue visual ETL :

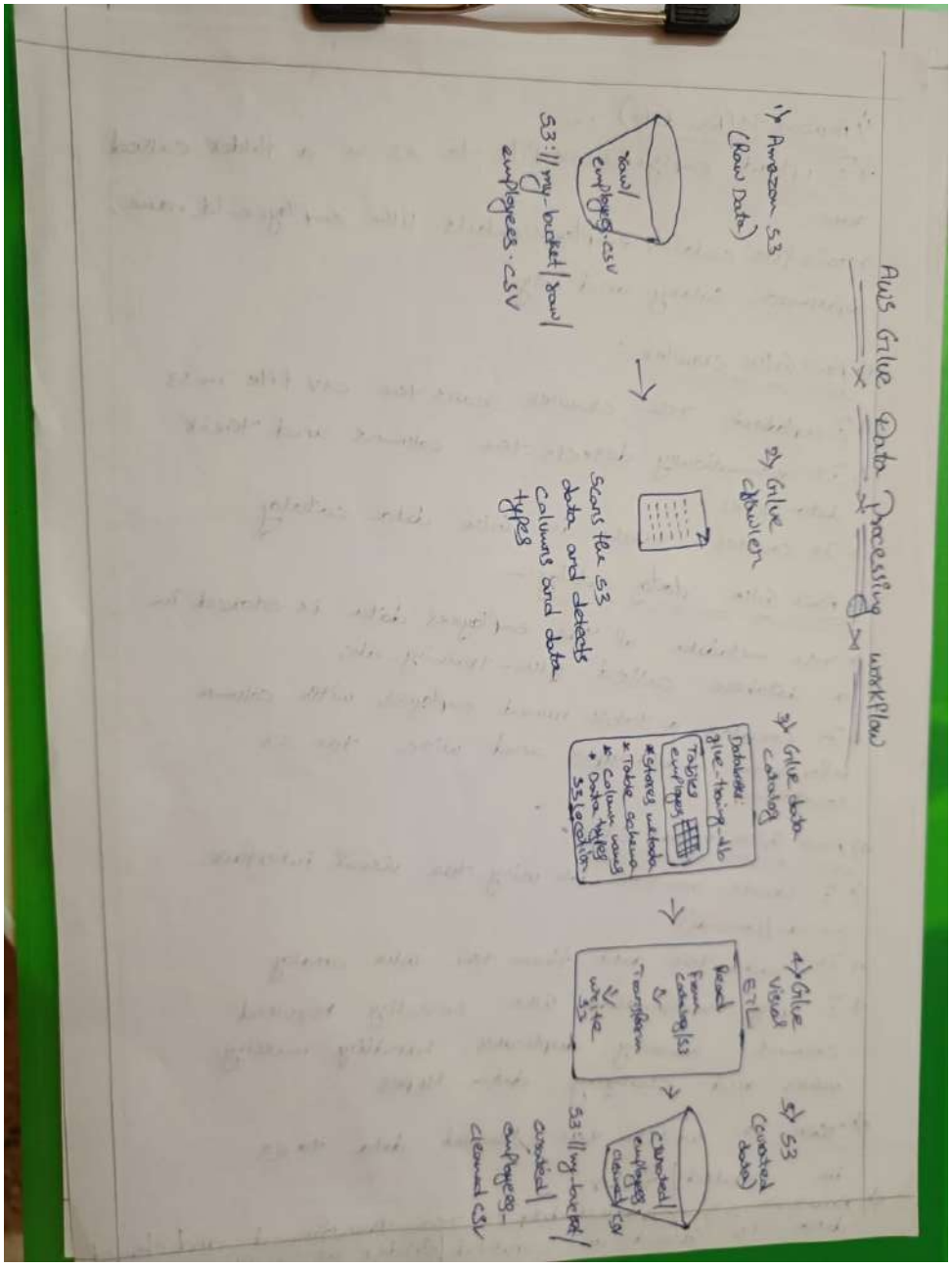
=> i) create an ETL job using the visual interface (no code / low code).

=> It reads the data from the Glue catalog

=> i) apply transformations like selecting required columns, removing duplicates, handling missing values and changing data types

=> Then it writes the cleaned data to S3 in curated/ folder

5) Amazon S3 (curated data) :- The transformed and cleaned data is saved in curated/folder as employees_cleaned



Final Explanation: AWS Glue Data Processing Workflow

1. First, I uploaded the `employees.csv` file to Amazon S3 and stored it inside the `raw/` folder.
2. I created an AWS IAM role to allow AWS Glue to access the required S3 data securely.
3. I created a Glue Crawler to scan the CSV file and identify its columns and data types.
4. The crawler stored the detected table information in the AWS Glue Data Catalog database named `glue_training_db`.
5. The Data Catalog helps me organize and manage the metadata of my employee data.
6. Next, I use AWS Glue Visual ETL to create a data processing workflow without writing much code.
7. During transformation, I can clean the data, remove duplicates, handle missing values, and select the required columns.
8. After processing, I configure the workflow to save the cleaned employee data in the `curated/` folder in Amazon S3.
9. Finally, the curated data can be used for analysis and reporting with services such as Amazon Athena.